

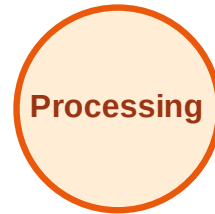
# BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

## Legend



Complete



Processing

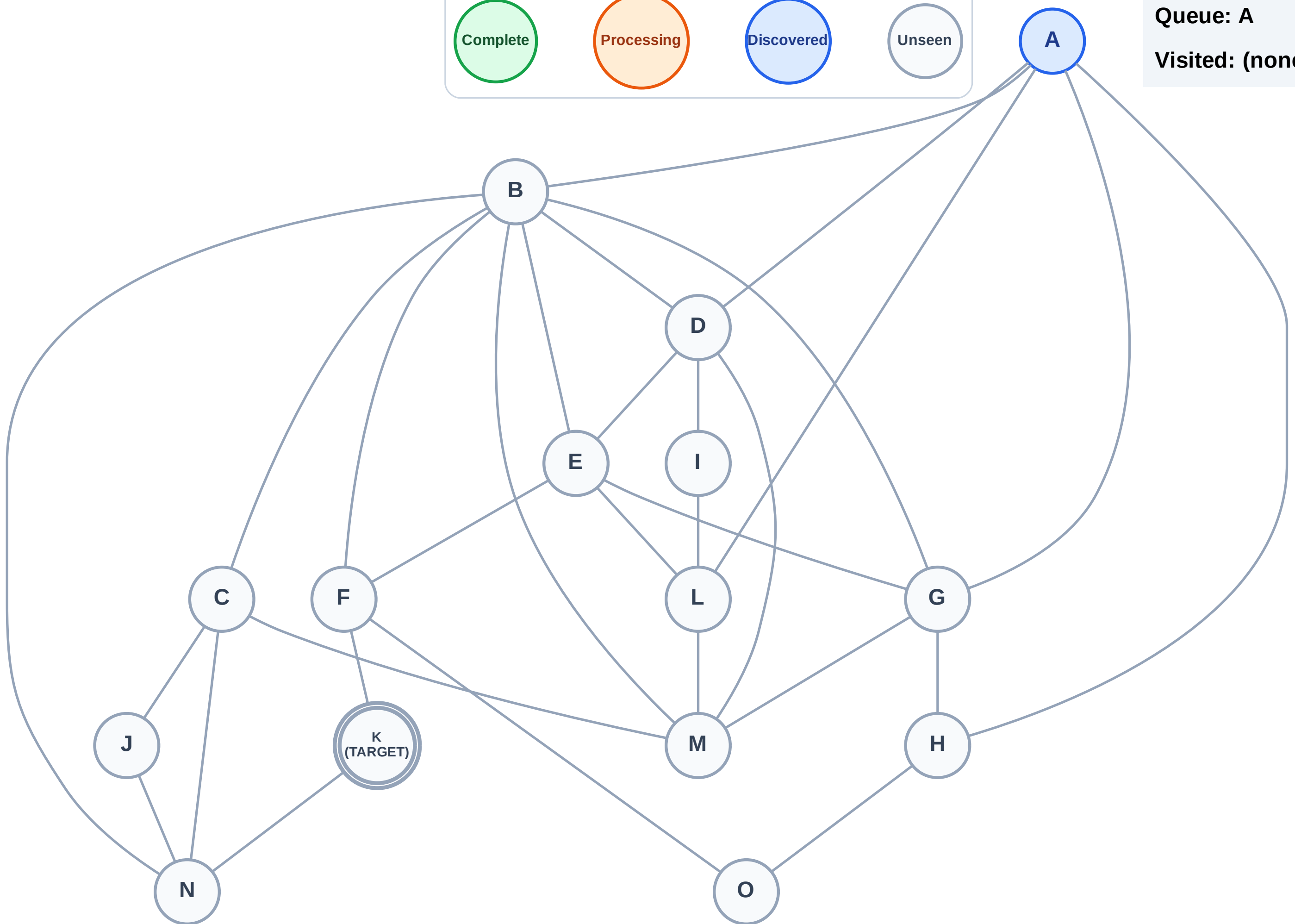


Discovered



Unseen

Queue: A  
Visited: (none)



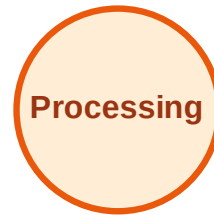
# BFS Traversal

Step 2: Process node A

## Legend



Complete



Processing



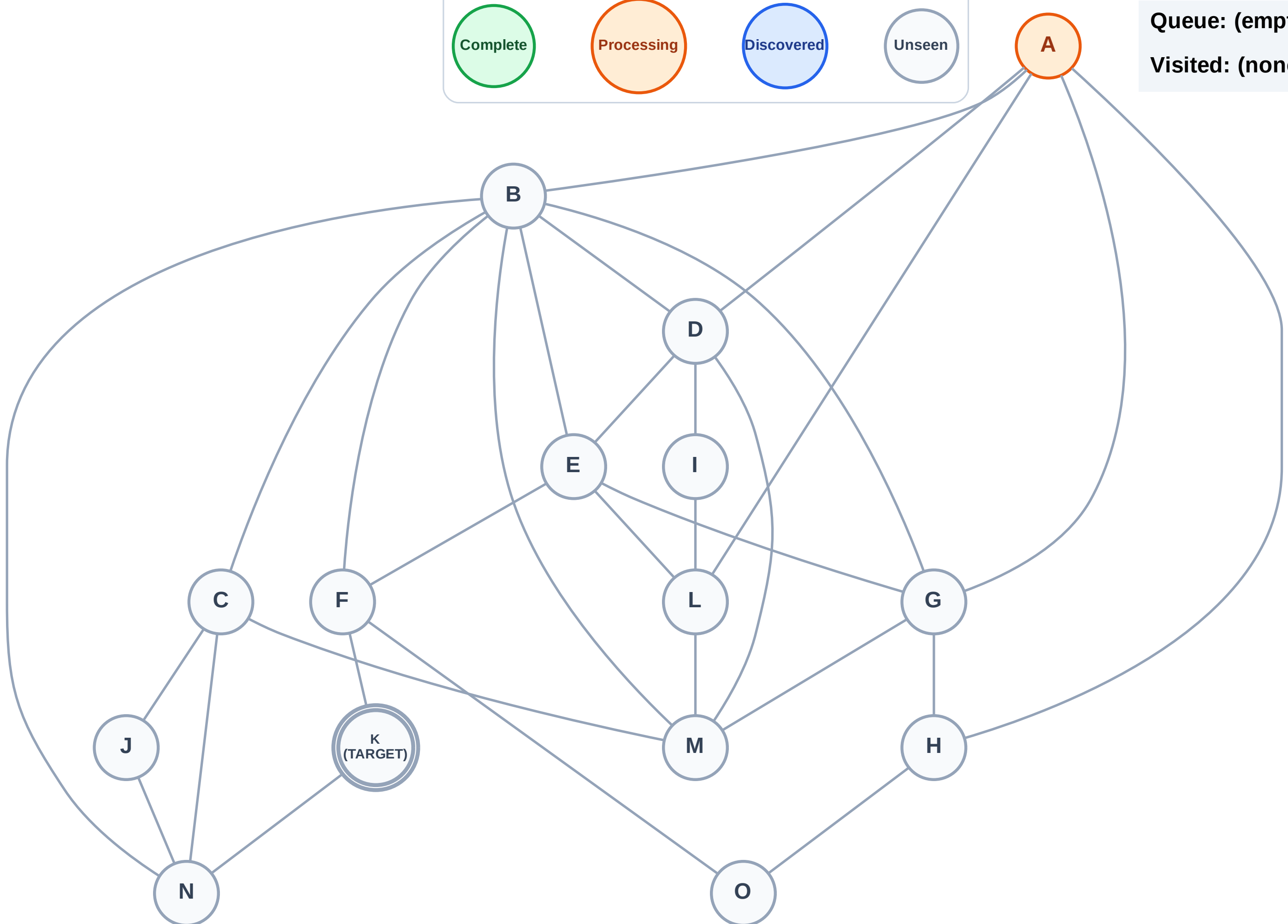
Discovered



Unseen

Queue: (empty)

Visited: (none)



# BFS Traversal

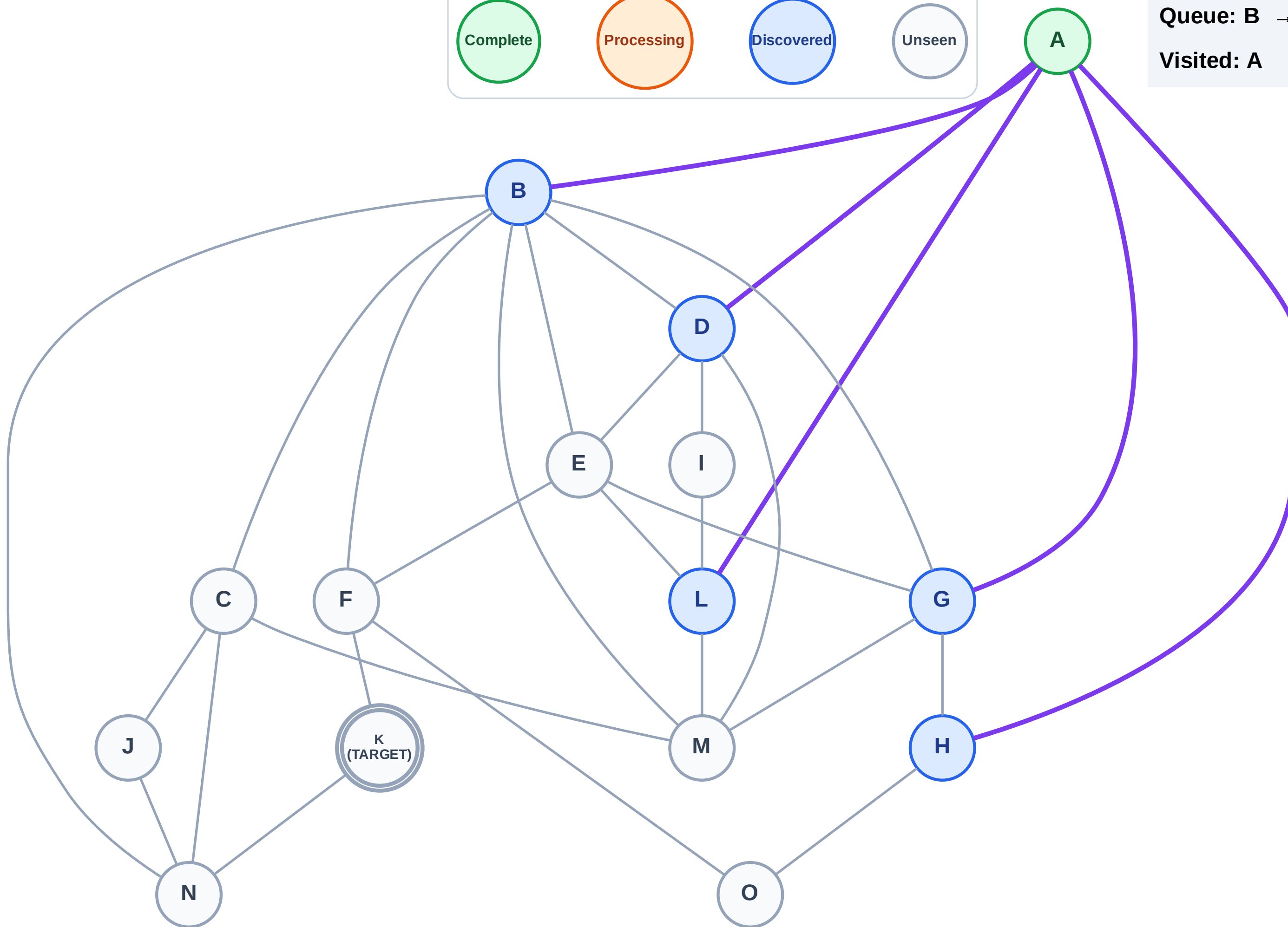
Step 3: Discover B, D, G, H, L from A

## Legend



Queue: B → D → G → H → L

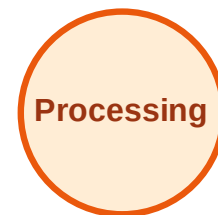
Visited: A



# BFS Traversal

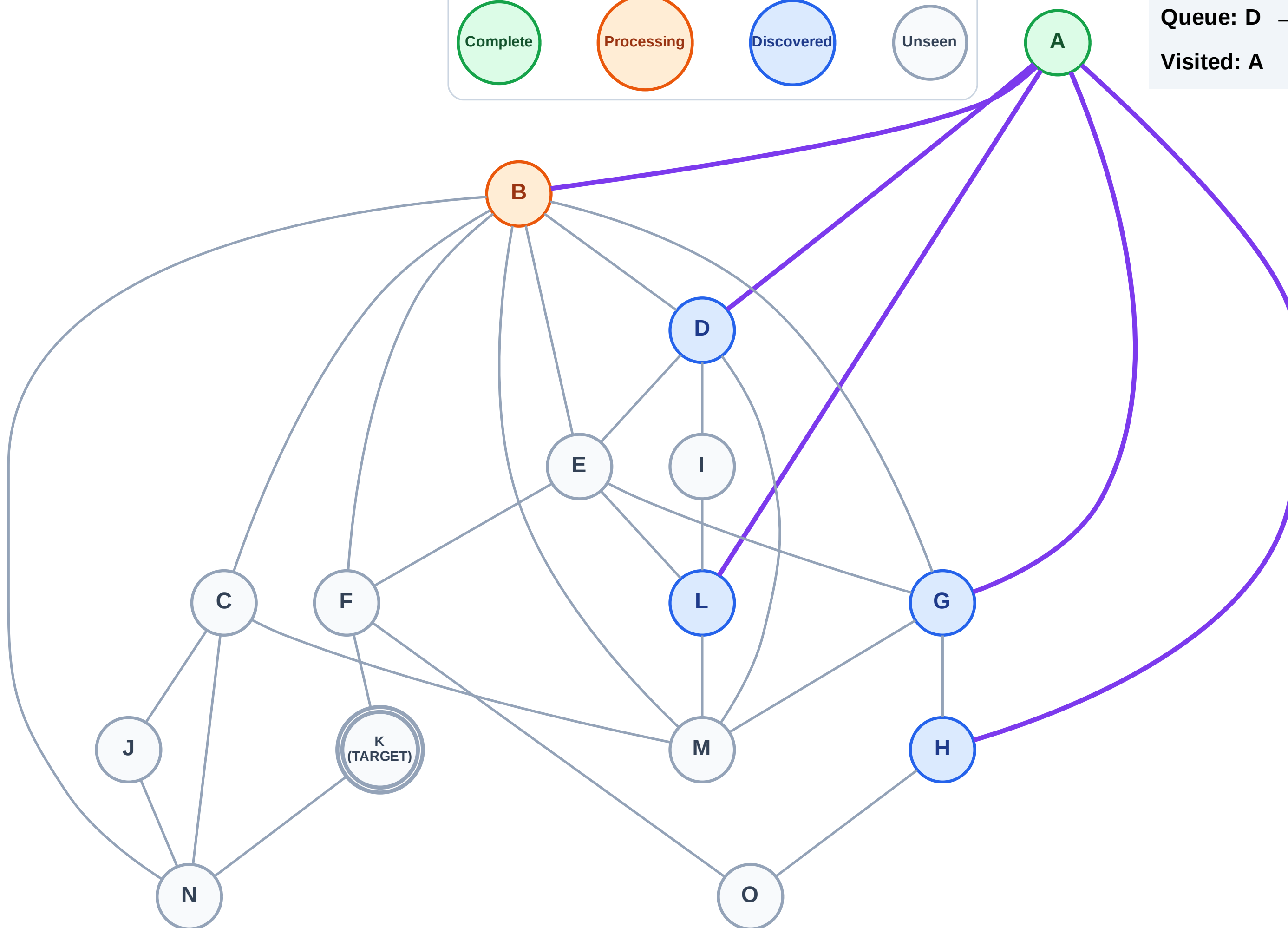
Step 4: Process node B

## Legend



Queue: D → G → H → L

Visited: A



# BFS Traversal

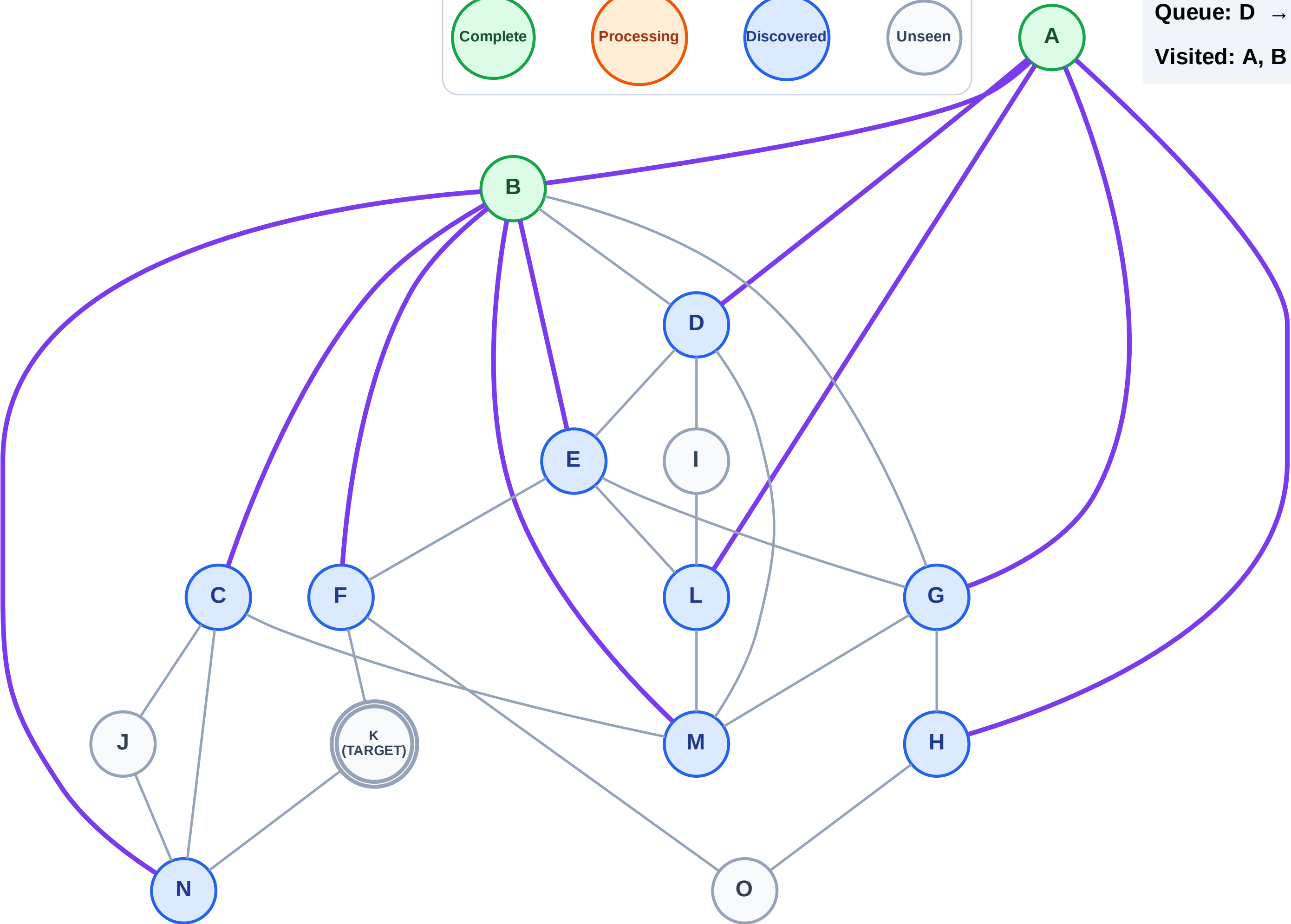
Step 5: Discover C, E, F, M, N from B

## Legend



Queue: D → G → H → L → C → E → F → M → N

Visited: A, B



# BFS Traversal

Step 6: Process node D

Legend

Complete

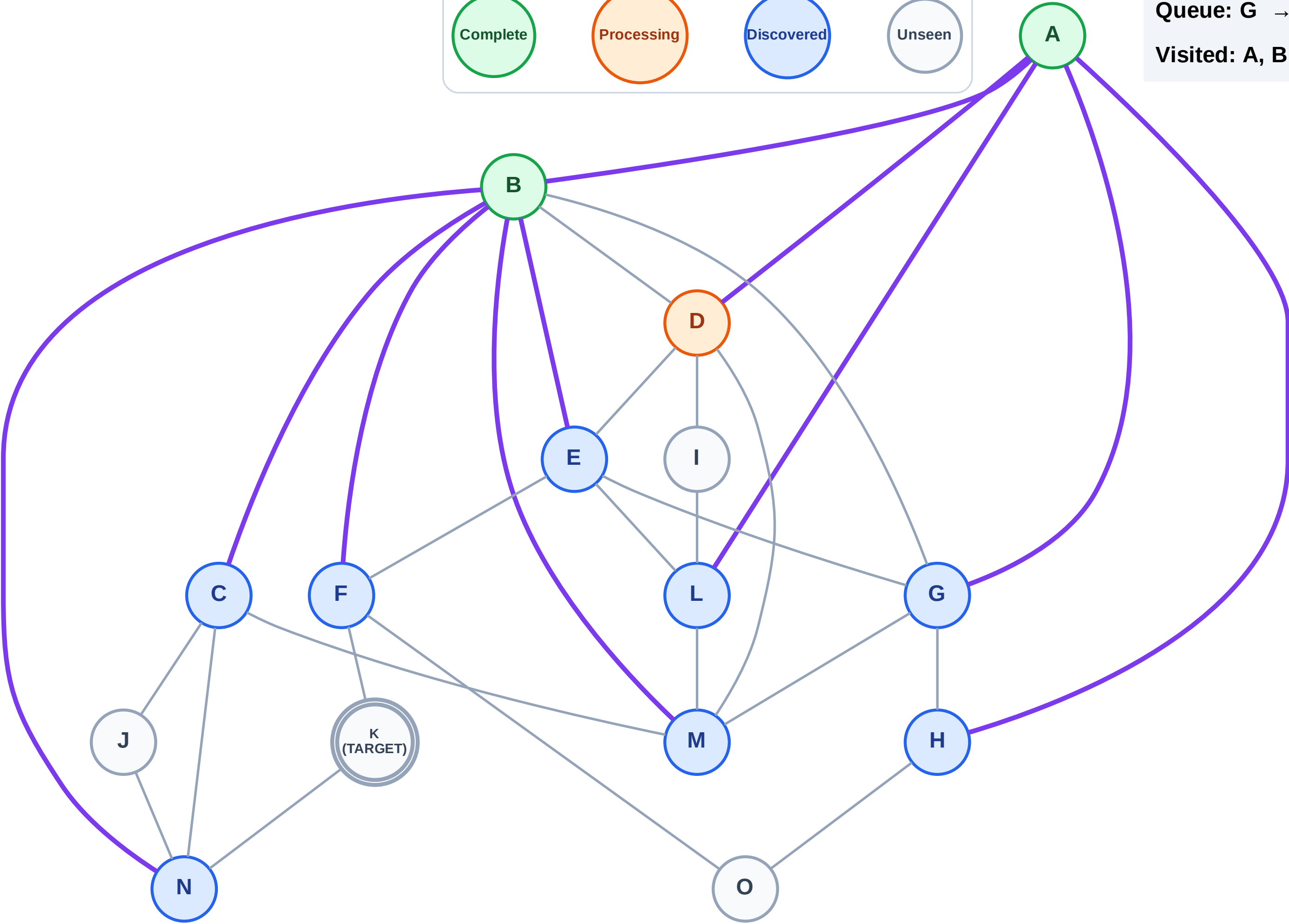
Processing

Discovered

Unseen

Queue: G → H → L → C → E → F → M → N

Visited: A, B



# BFS Traversal

Step 7: Discover I from D

## Legend



Complete



Processing



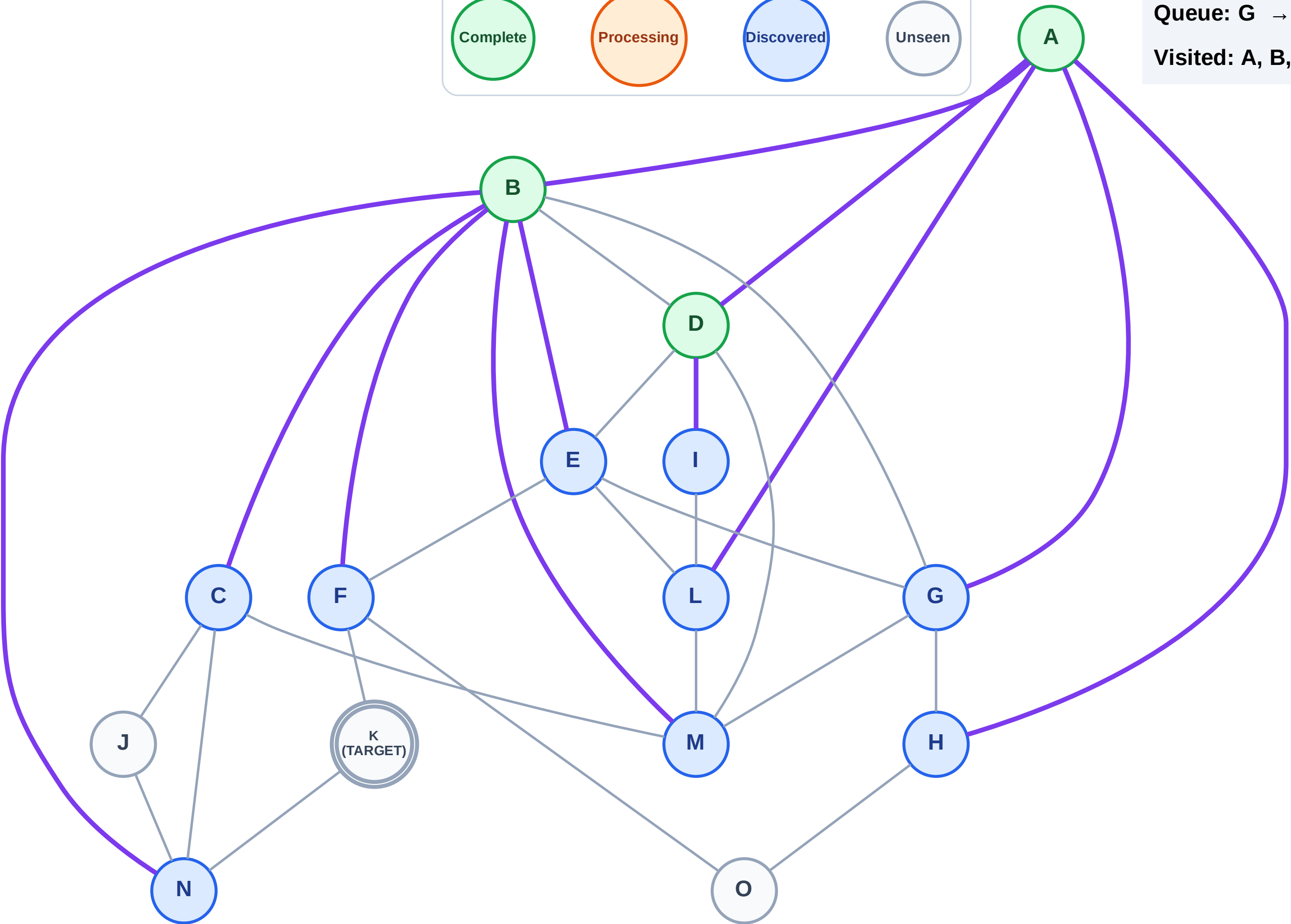
Discovered



Unseen

Queue: G → H → L → C → E → F → M → N → I

Visited: A, B, D



# BFS Traversal

Step 8: Process node G

Legend

Complete

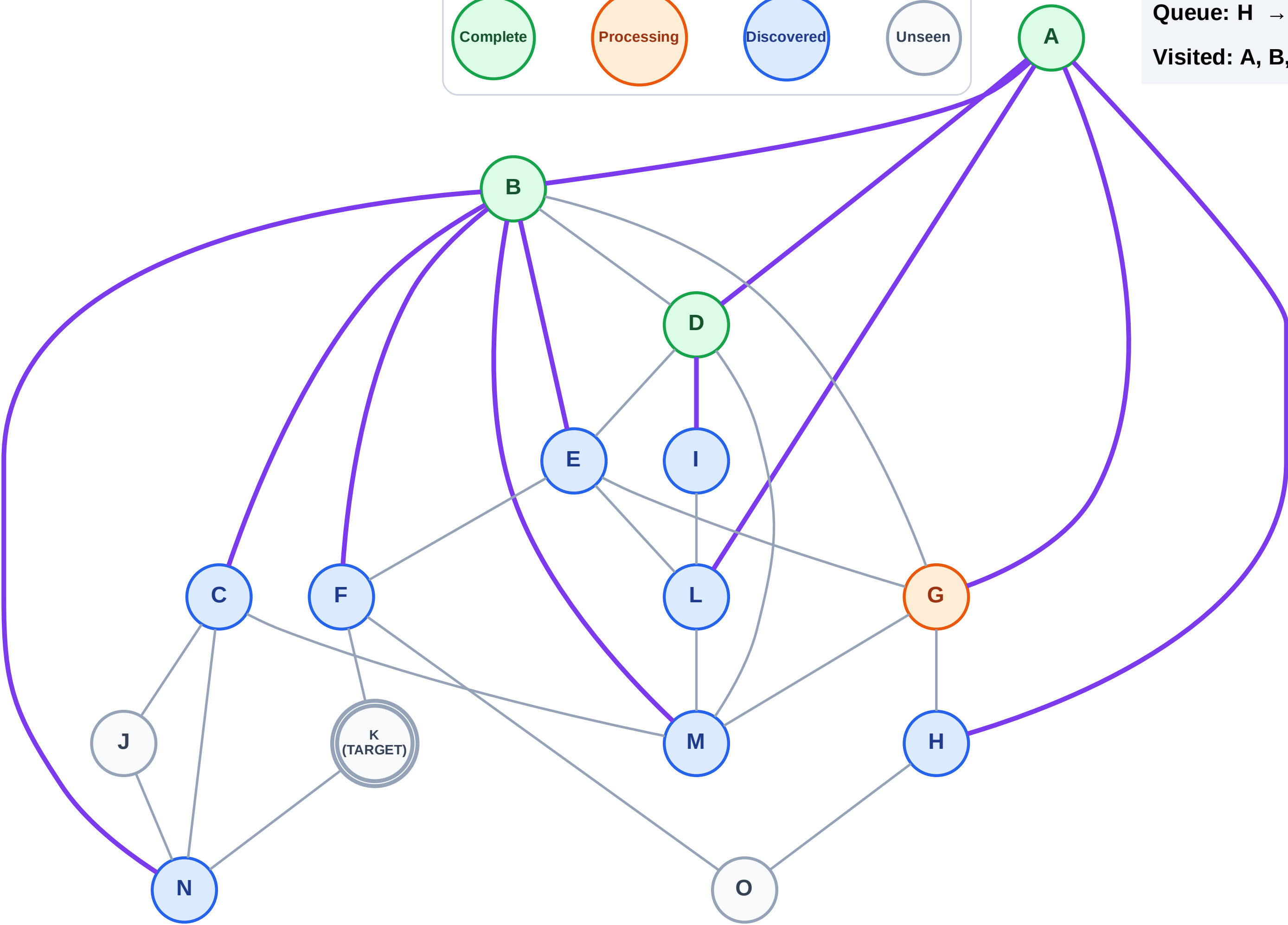
Processing

Discovered

Unseen

Queue: H → L → C → E → F → M → N → I

Visited: A, B, D



# BFS Traversal

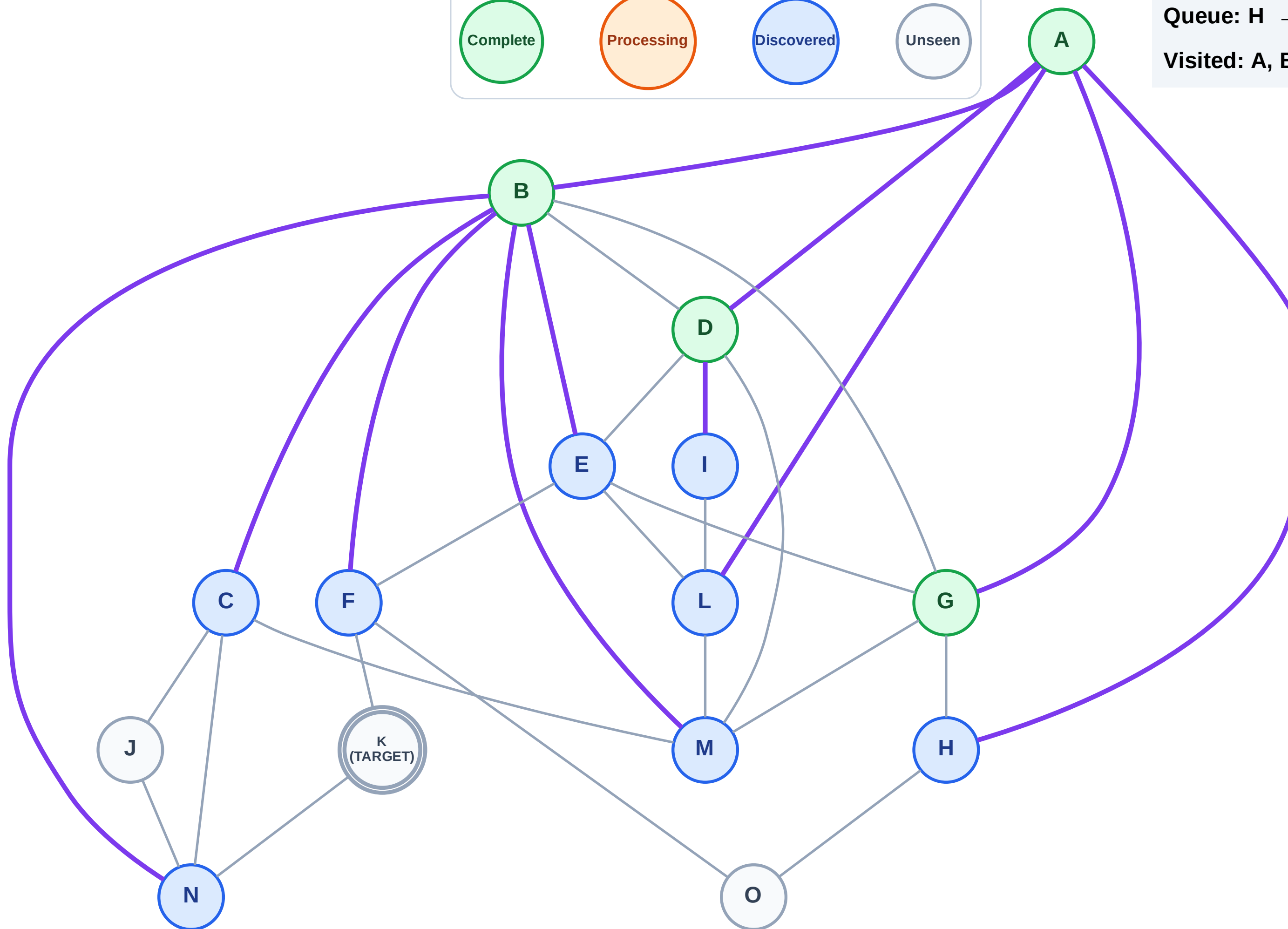
Step 9: No new neighbors from G

## Legend



Queue: H → L → C → E → F → M → N → I

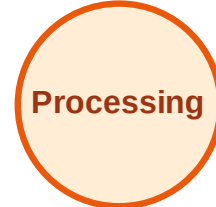
Visited: A, B, D, G



# BFS Traversal

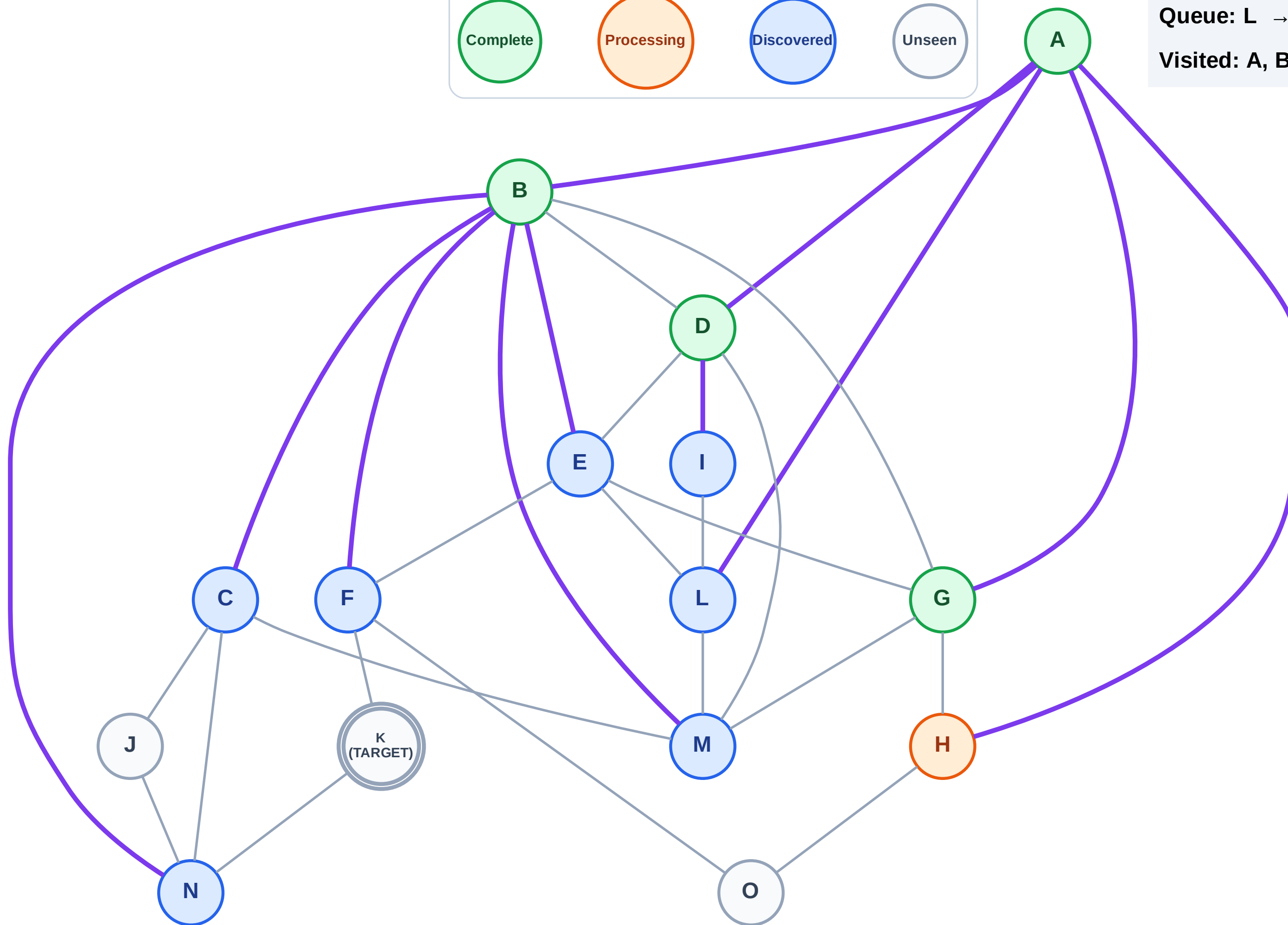
Step 10: Process node H

## Legend



Queue: L → C → E → F → M → N → I

Visited: A, B, D, G



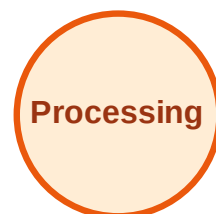
# BFS Traversal

Step 11: Discover O from H

## Legend



Complete



Processing



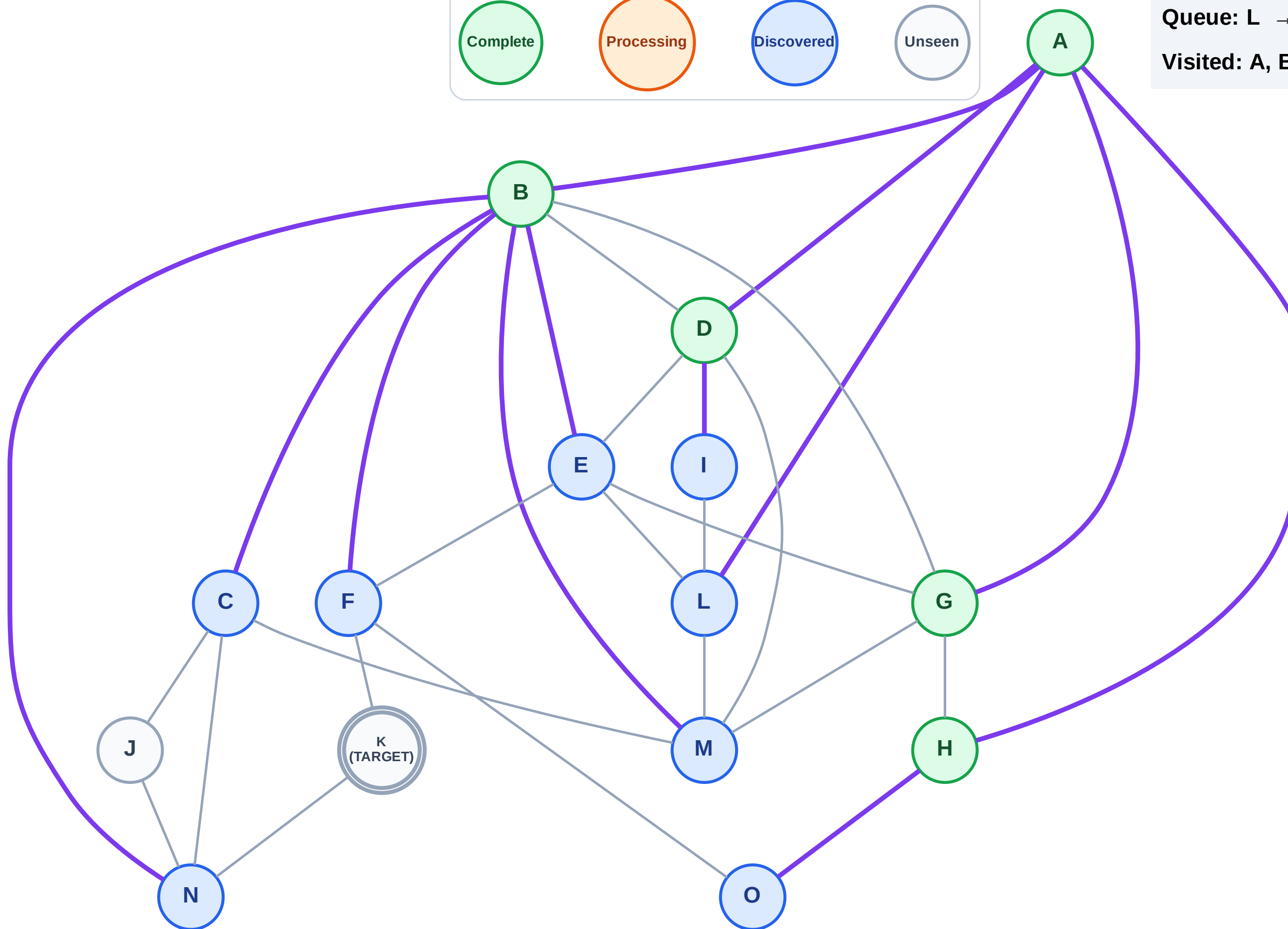
Discovered



Unseen

Queue: L → C → E → F → M → N → I → O

Visited: A, B, D, G, H



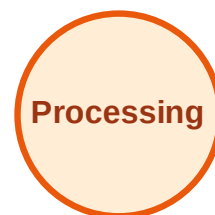
# BFS Traversal

Step 12: Process node L

## Legend



Complete



Processing

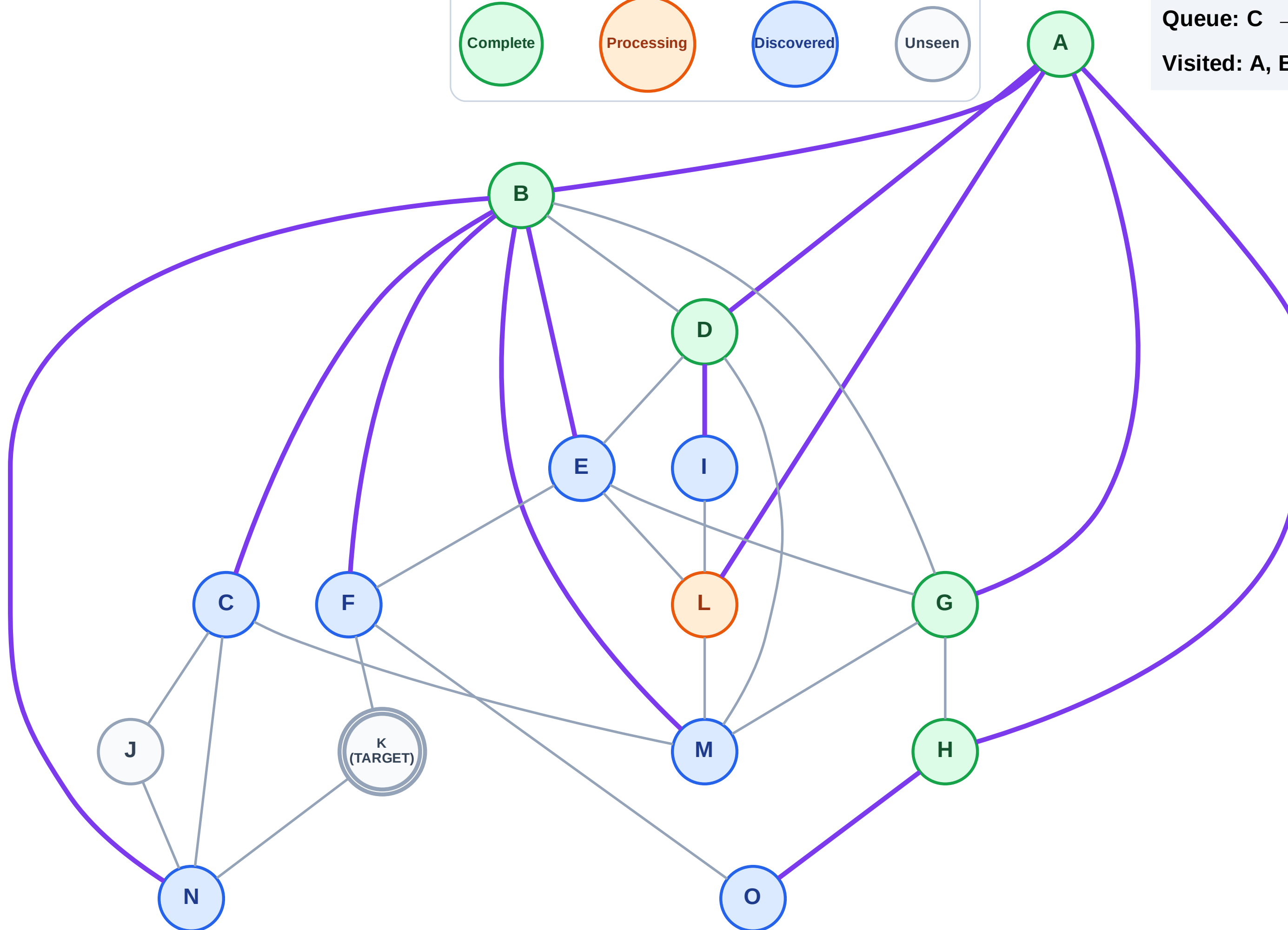


Discovered



Unseen

Queue: C → E → F → M → N → I → O  
Visited: A, B, D, G, H



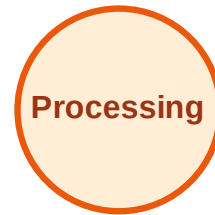
# BFS Traversal

Step 13: No new neighbors from L

## Legend



Complete



Processing

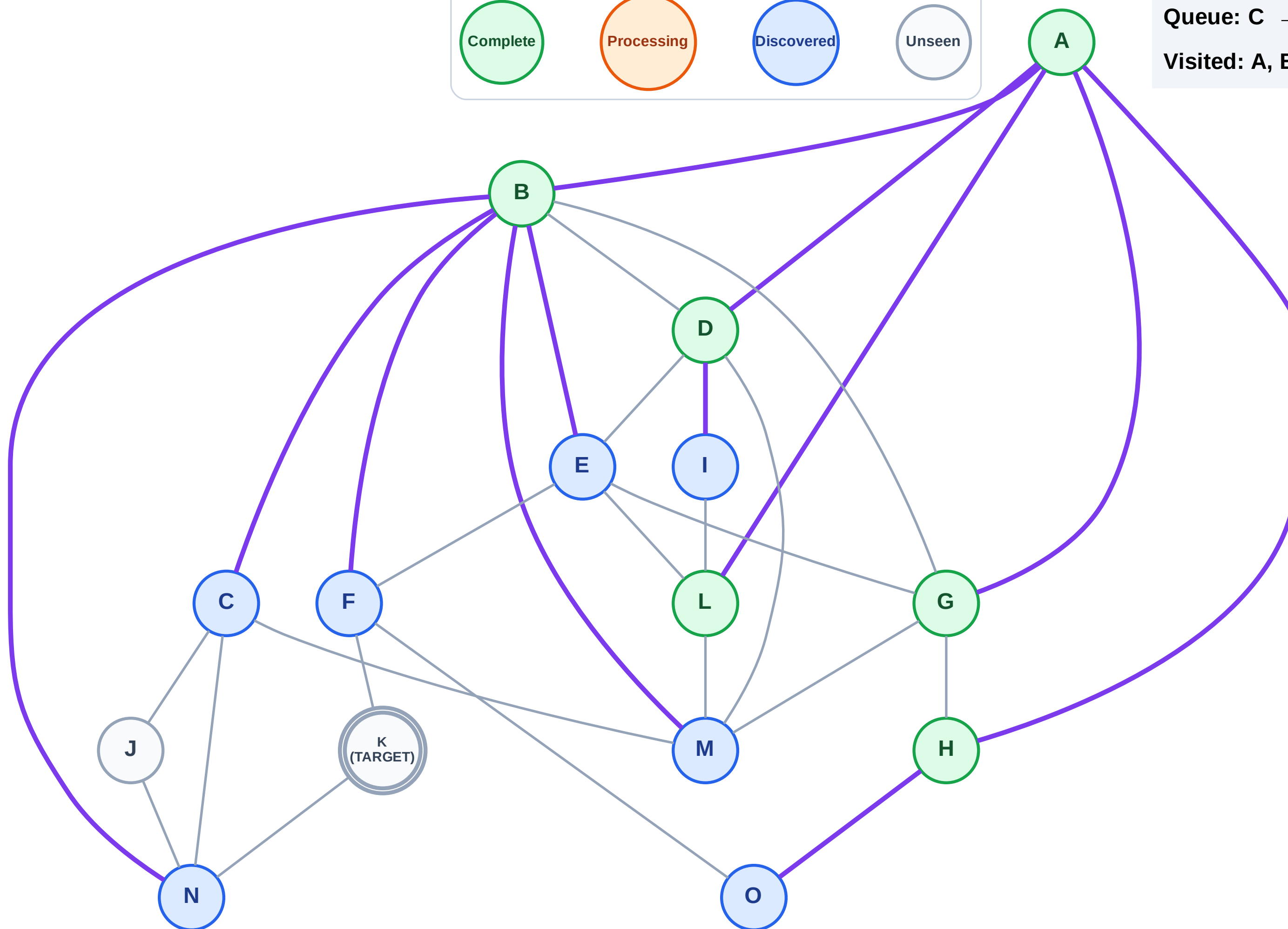


Discovered



Unseen

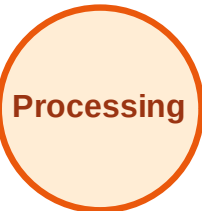
Queue: C → E → F → M → N → I → O  
Visited: A, B, D, G, H, L



# BFS Traversal

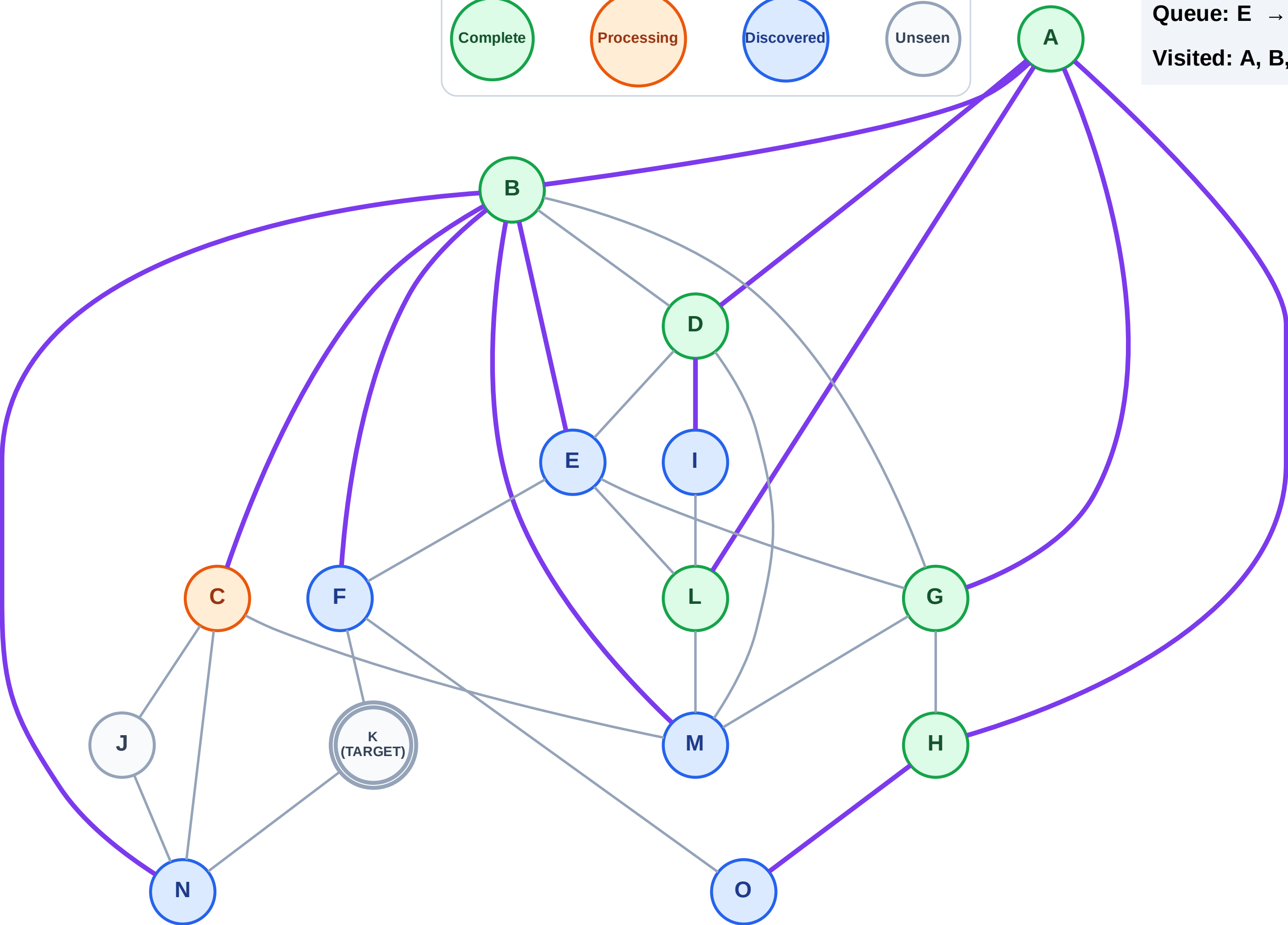
Step 14: Process node C

## Legend



Queue: E → F → M → N → I → O

Visited: A, B, D, G, H, L



# BFS Traversal

Step 15: Discover J from C

Legend

Complete

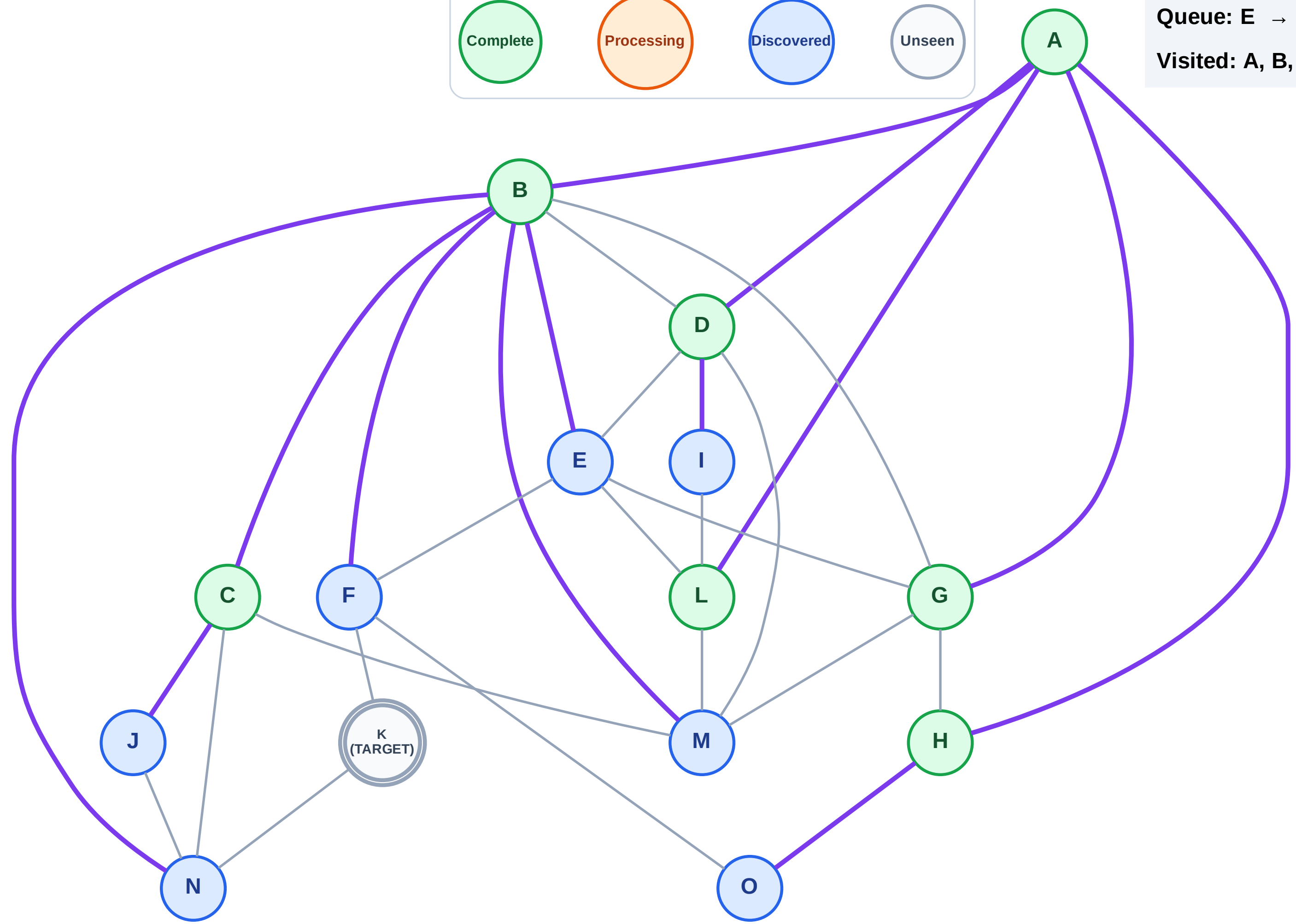
Processing

Discovered

Unseen

Queue: E → F → M → N → I → O → J

Visited: A, B, C, D, G, H, L



# BFS Traversal

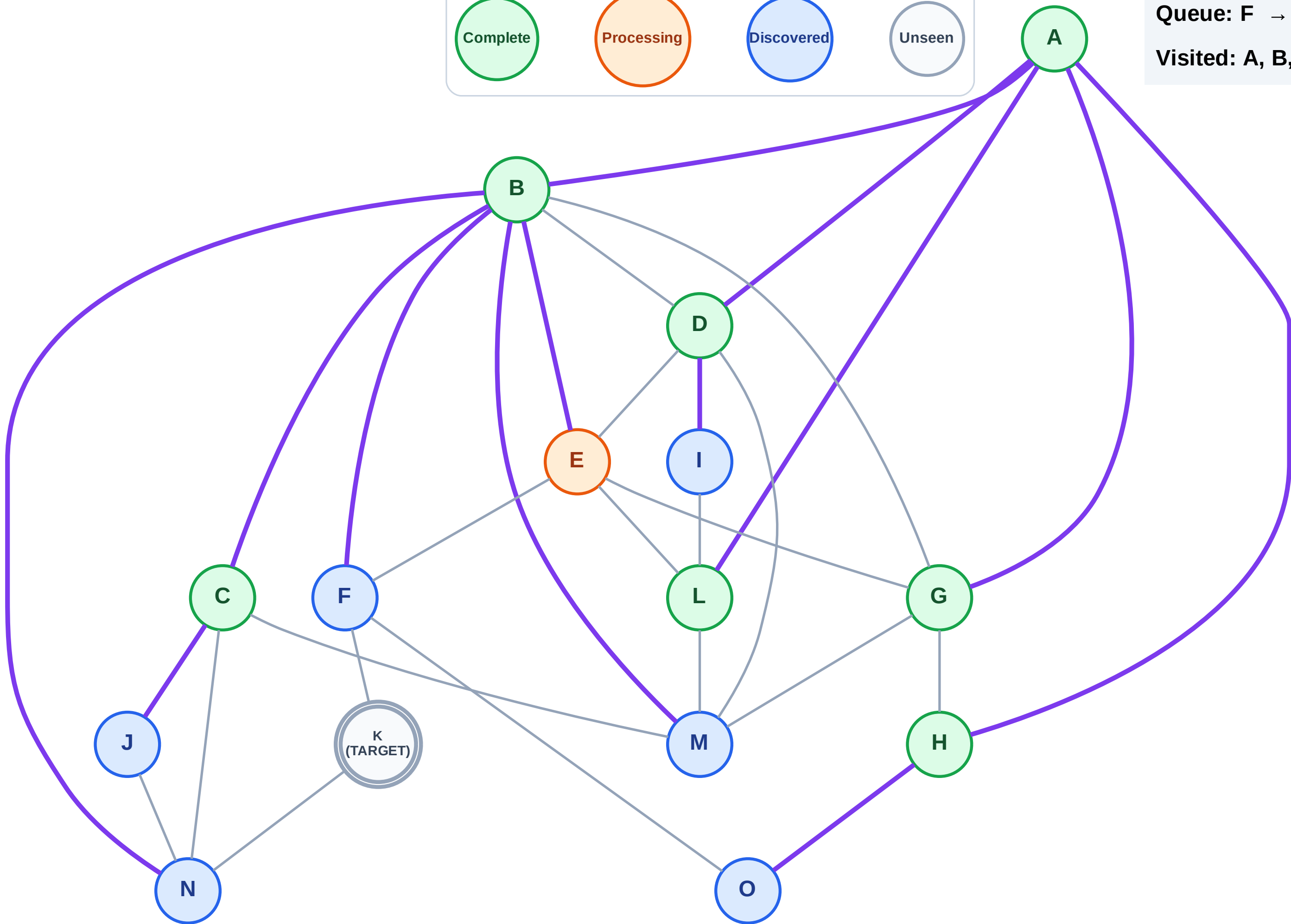
Step 16: Process node E

## Legend



Queue: F → M → N → I → O → J

Visited: A, B, C, D, G, H, L



# BFS Traversal

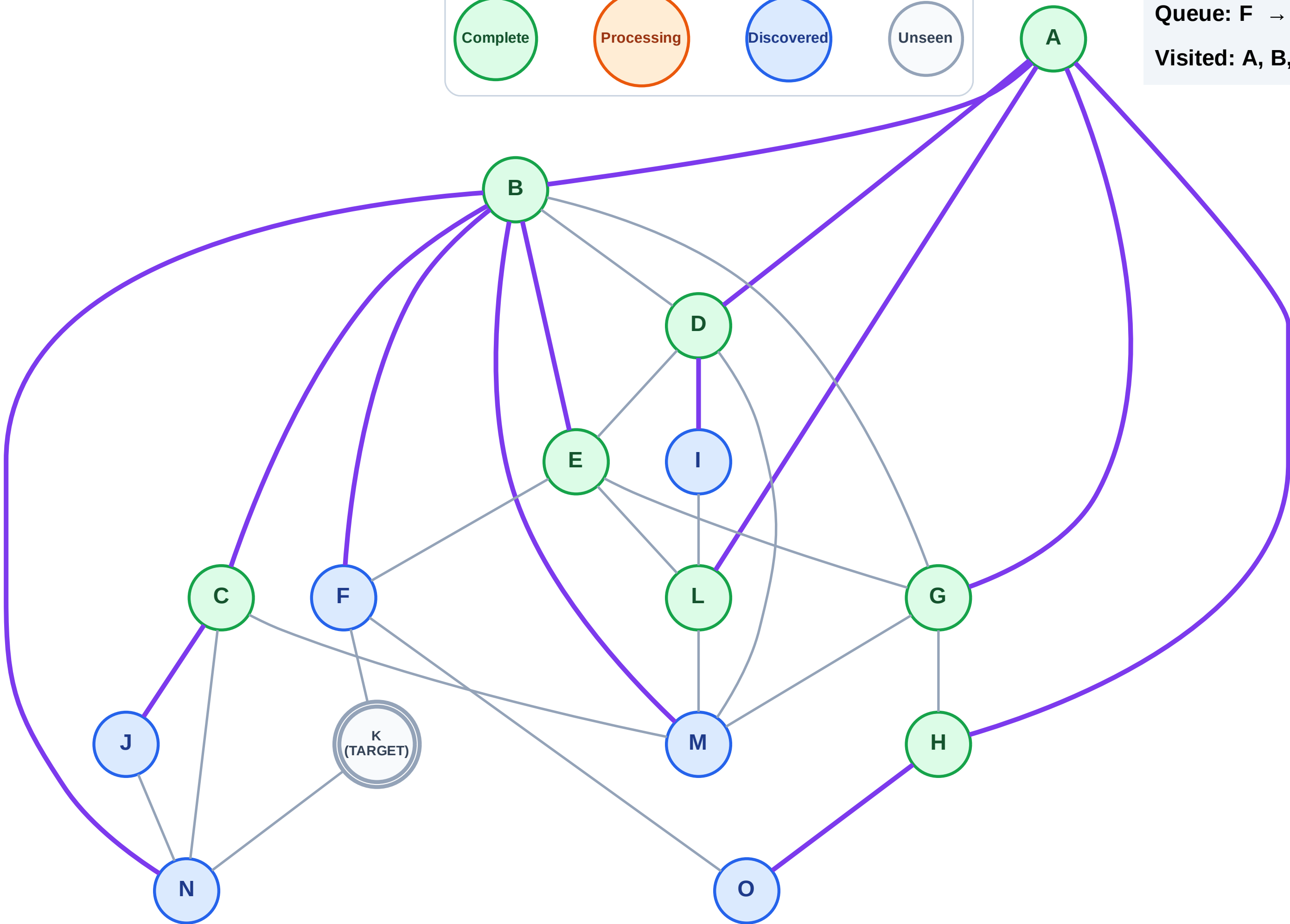
Step 17: No new neighbors from E

## Legend



Queue: F → M → N → I → O → J

Visited: A, B, C, D, E, G, H, L



# BFS Traversal

Step 18: Process node F

Legend

Complete

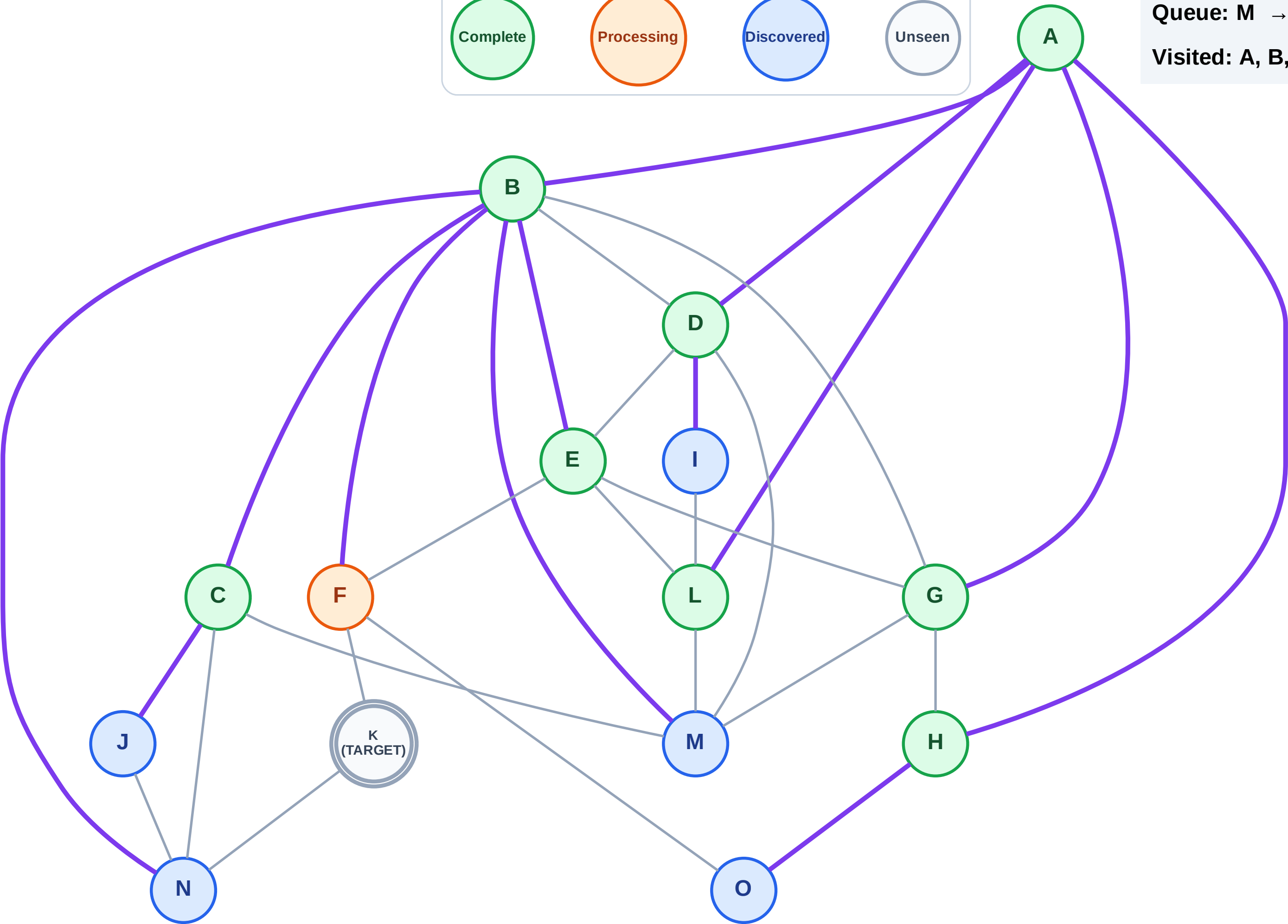
Processing

Discovered

Unseen

Queue: M → N → I → O → J

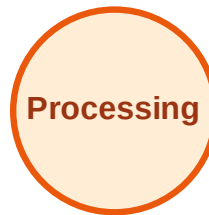
Visited: A, B, C, D, E, G, H, L



# BFS Traversal

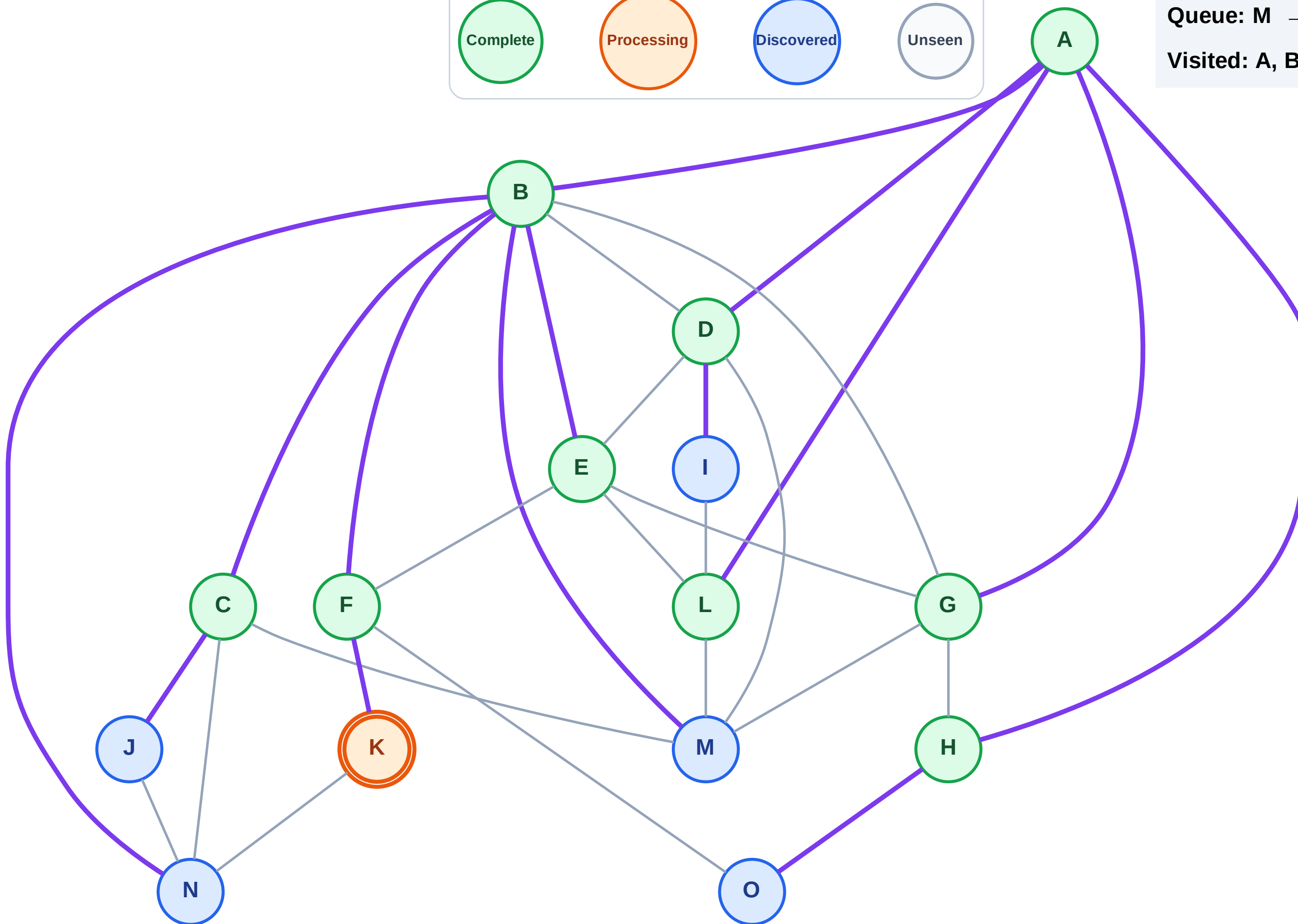
Step 19: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → F → K

## Legend



Queue: M → N → I → O → J

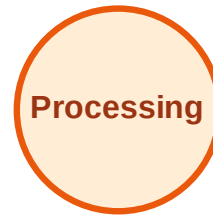
Visited: A, B, C, D, E, F, G, H, K, L



# DFS Traversal

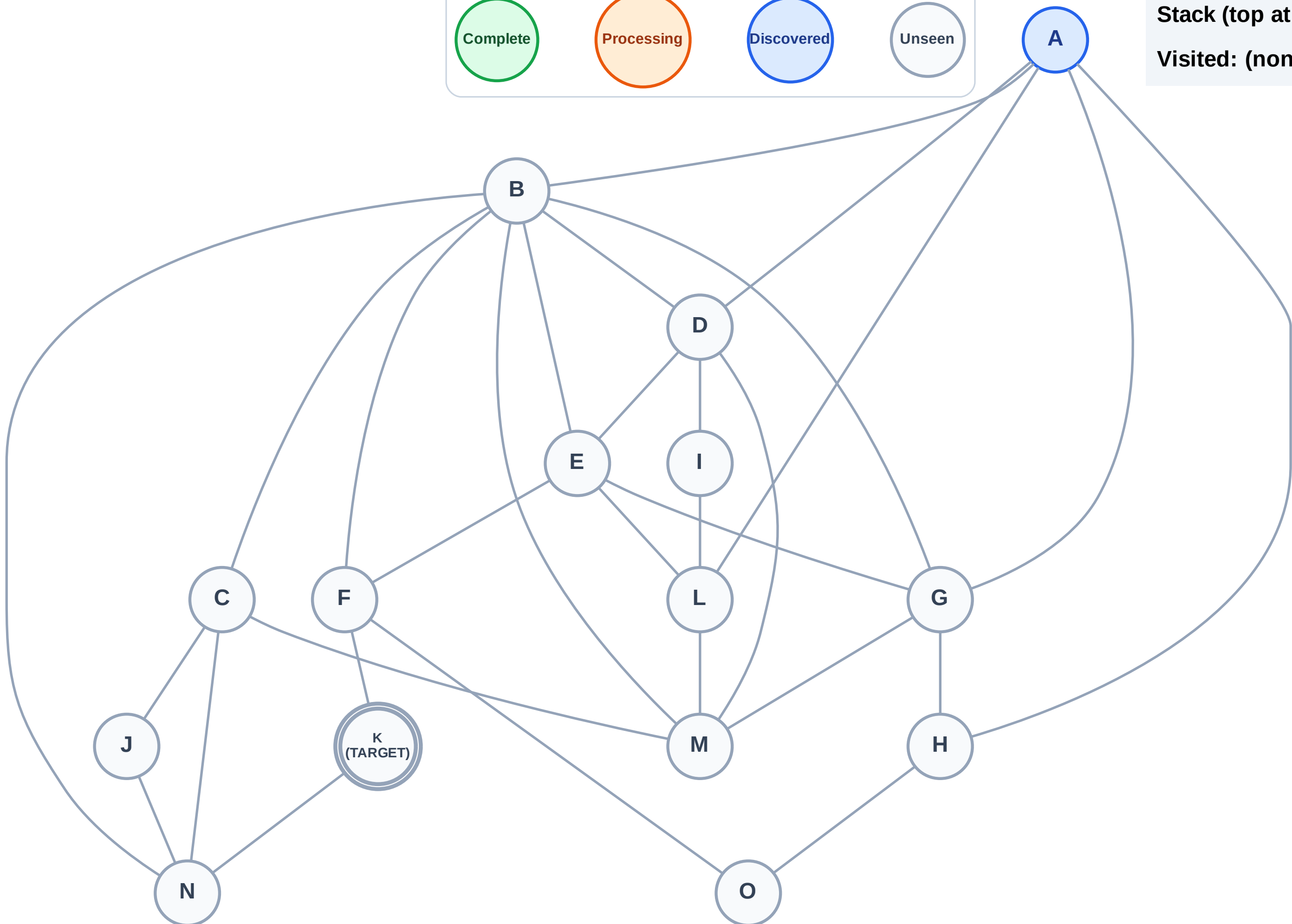
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

## Legend



Stack (top at right): A

Visited: (none)



# DFS Traversal

Step 2: Process node A

Legend

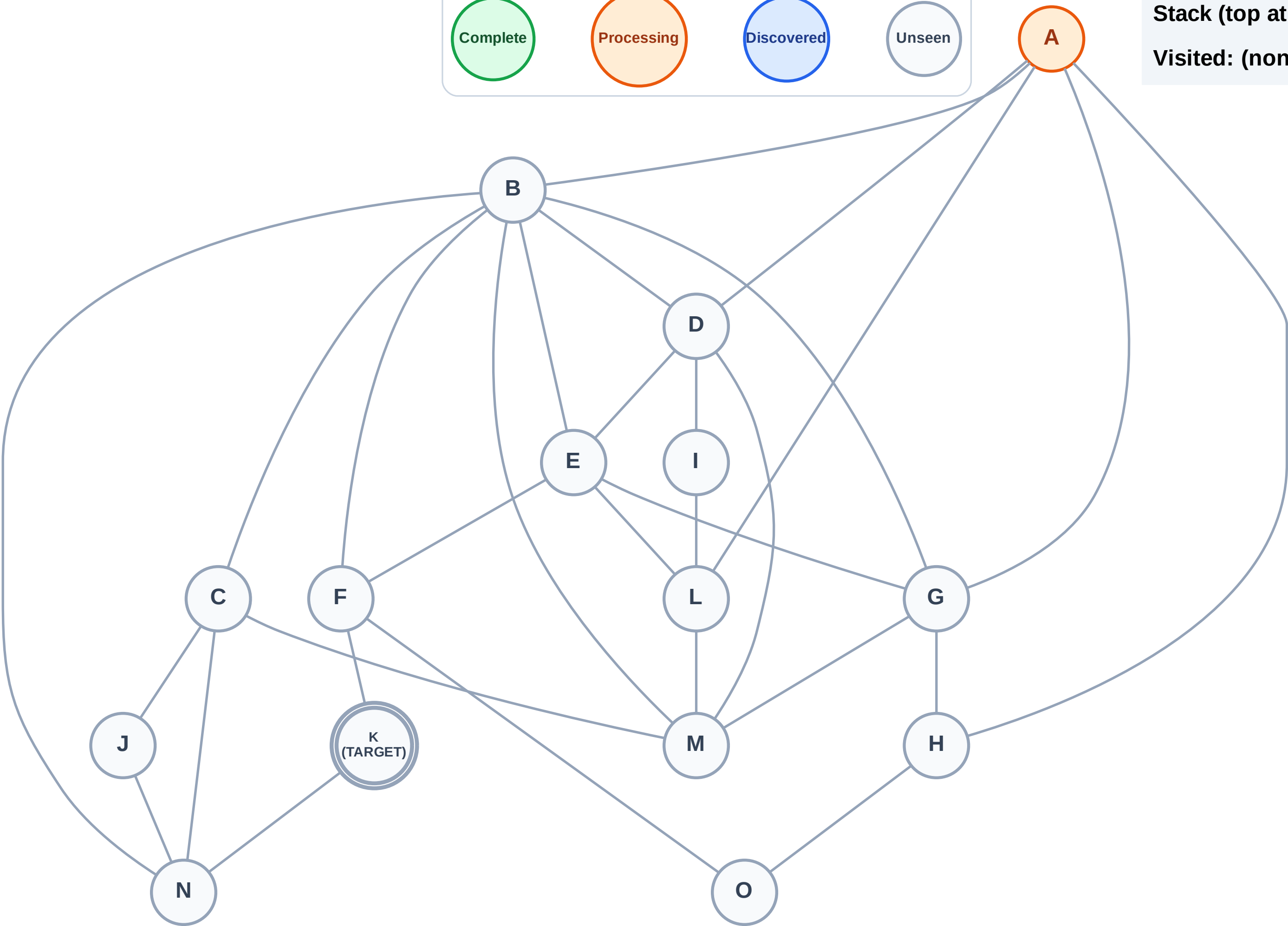
Complete

Processing

Discovered

Unseen

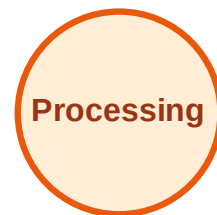
Stack (top at right): (empty)  
Visited: (none)



# DFS Traversal

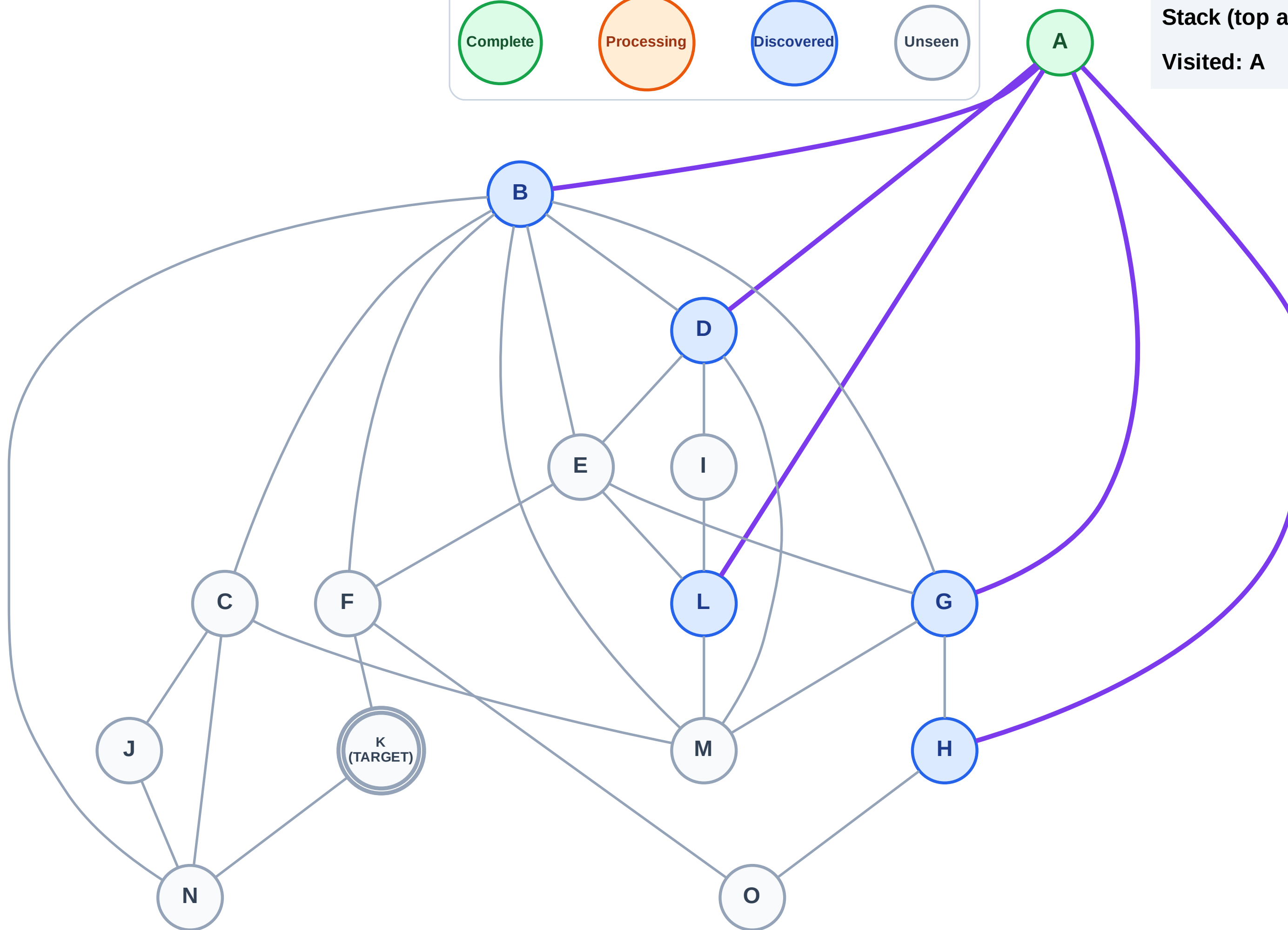
Step 3: Discover L, H, G, D, B from A

## Legend



Stack (top at right): L | H | G | D | B

Visited: A



# DFS Traversal

Step 4: Process node B

Legend

Complete

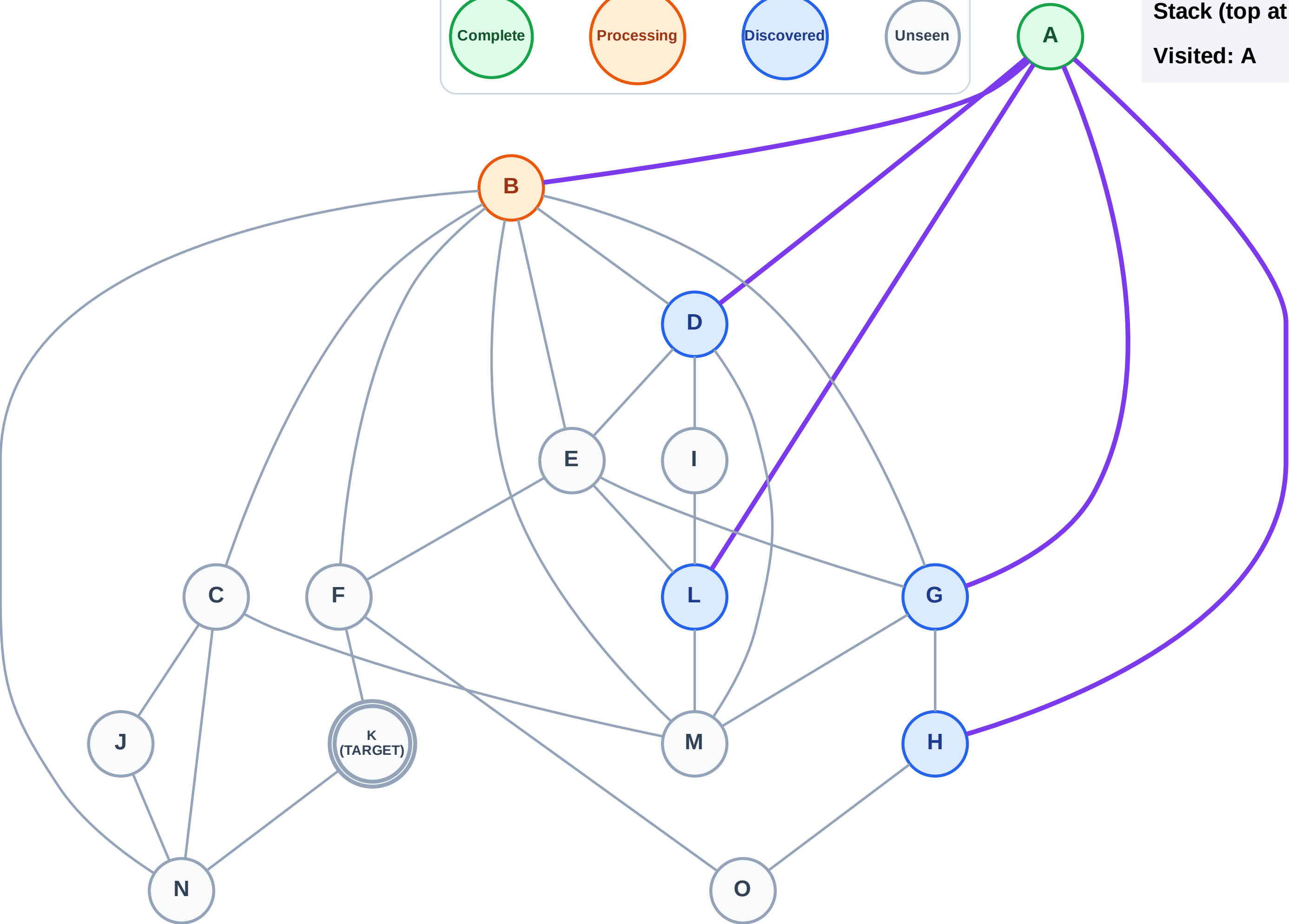
Processing

Discovered

Unseen

Stack (top at right): L | H | G | D

Visited: A



# DFS Traversal

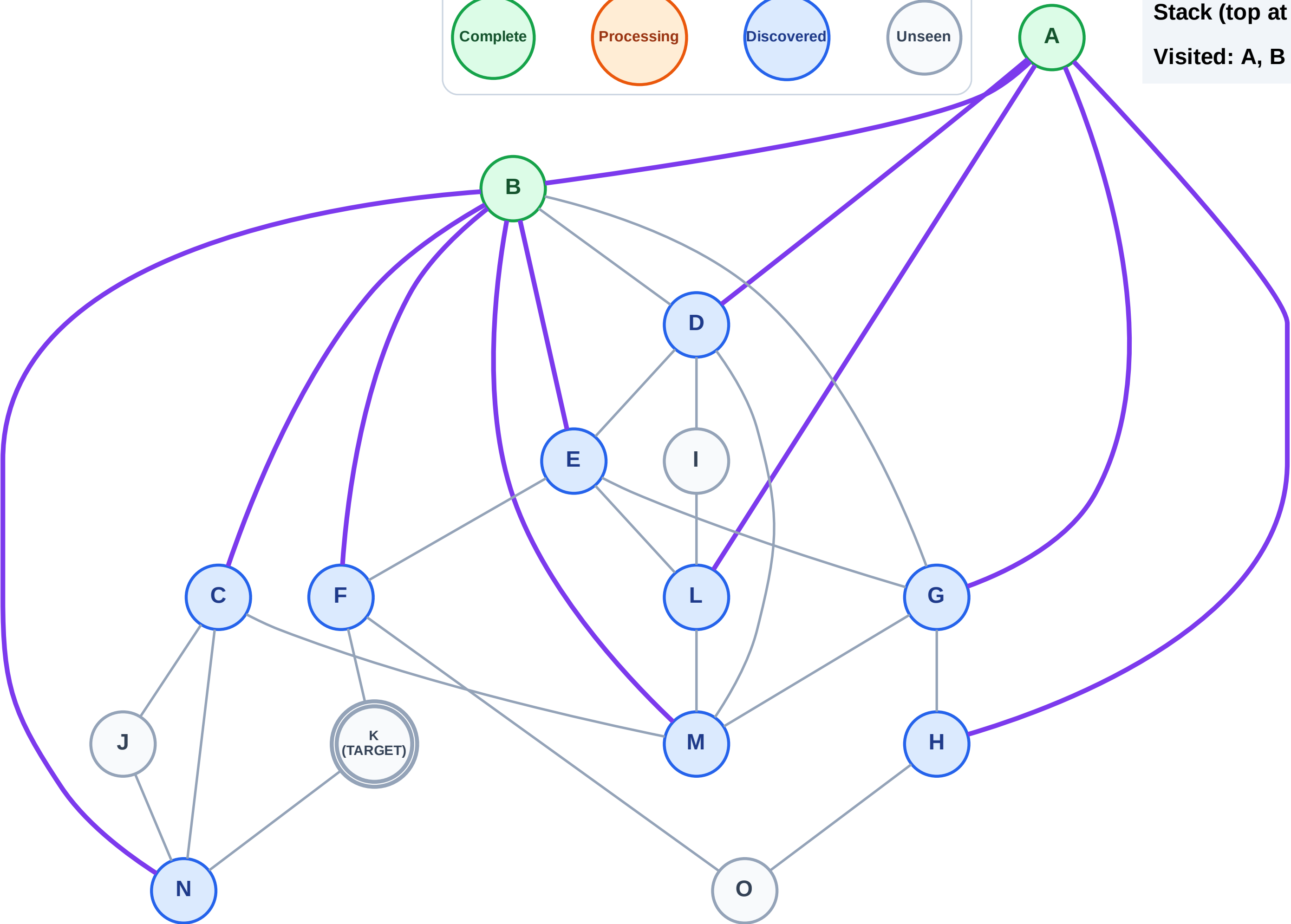
Step 5: Discover N, M, F, E, C from B

## Legend



Stack (top at right): L | H | G | D | N | M | F | E | C

Visited: A, B



# DFS Traversal

Step 6: Process node C

Legend

Complete

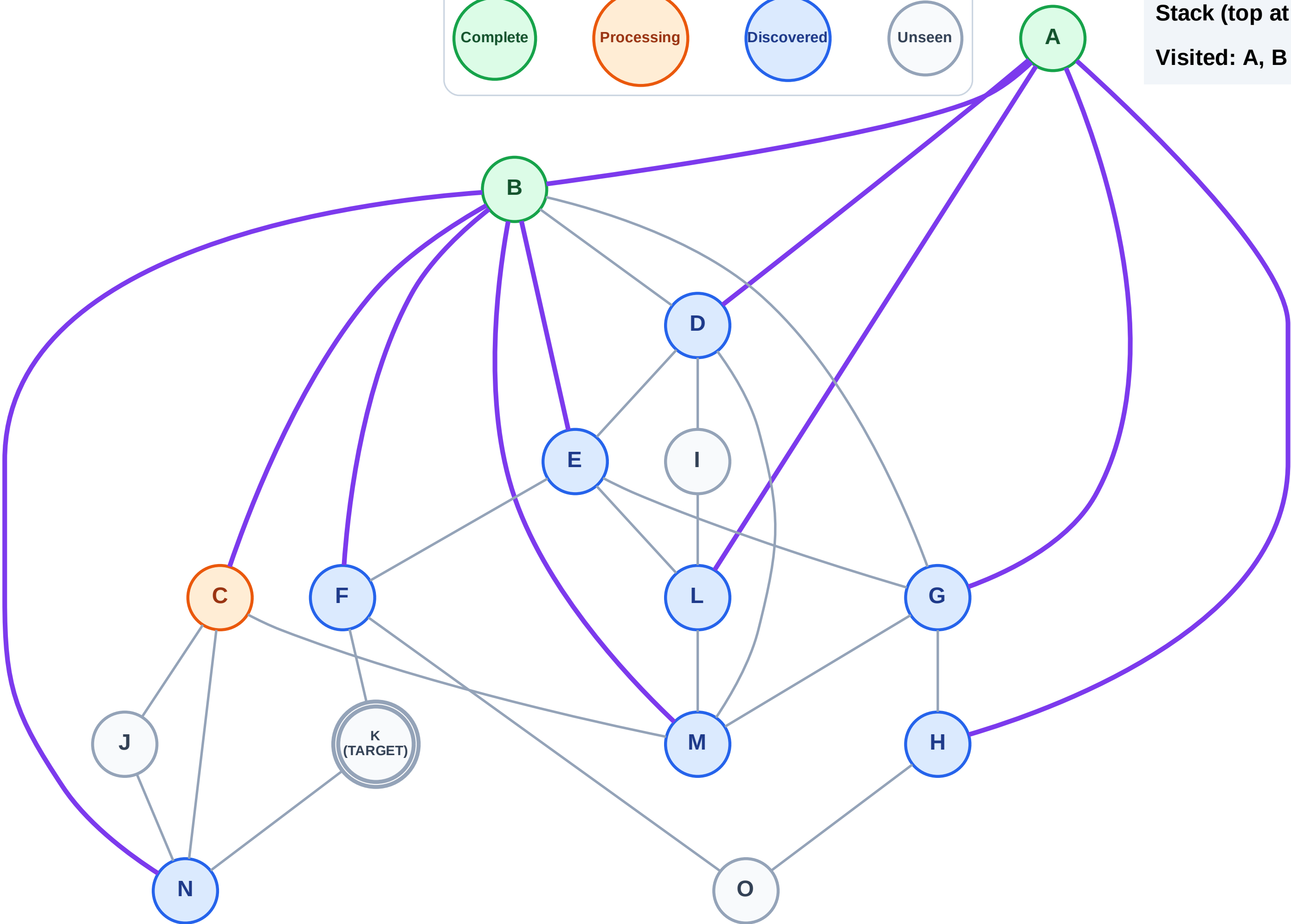
Processing

Discovered

Unseen

Stack (top at right): L | H | G | D | N | M | F | E

Visited: A, B



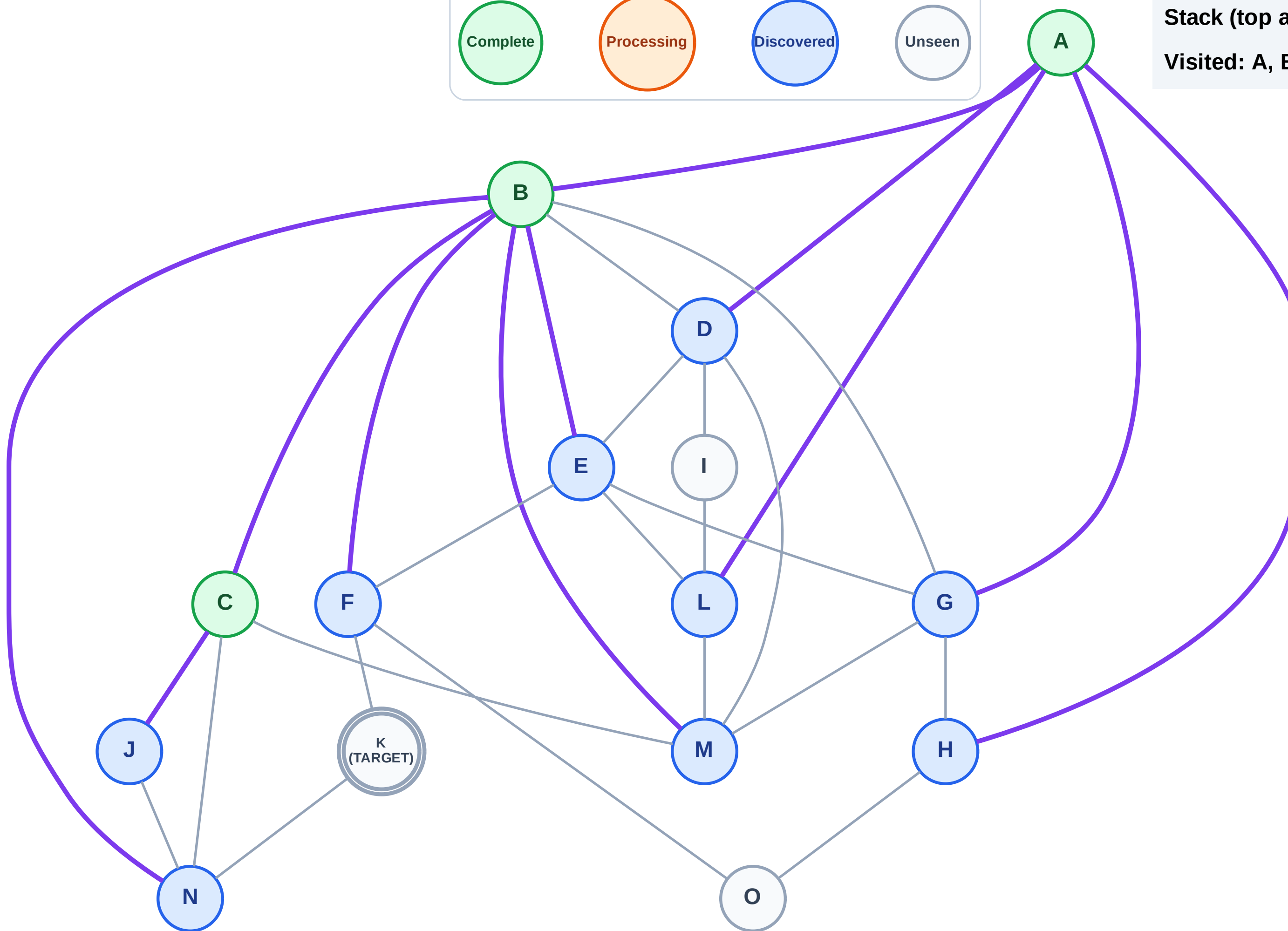
# DFS Traversal

Step 7: Discover J from C

## Legend



Stack (top at right): L | H | G | D | N | M | F | E | J  
Visited: A, B, C

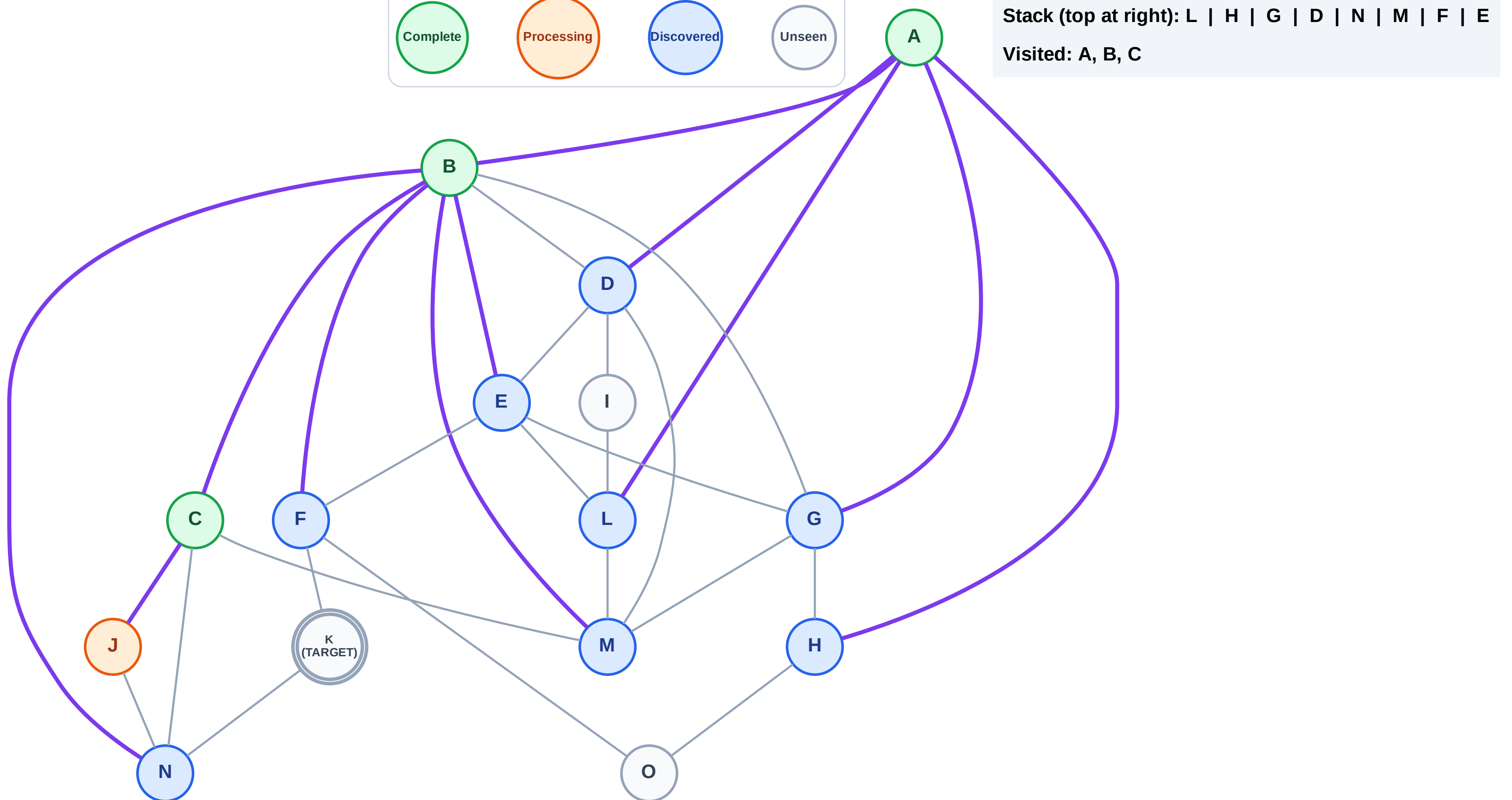


Step 8: Process node J

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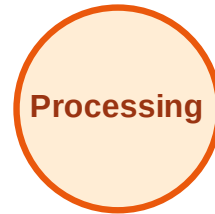
Visited: A B C



# DFS Traversal

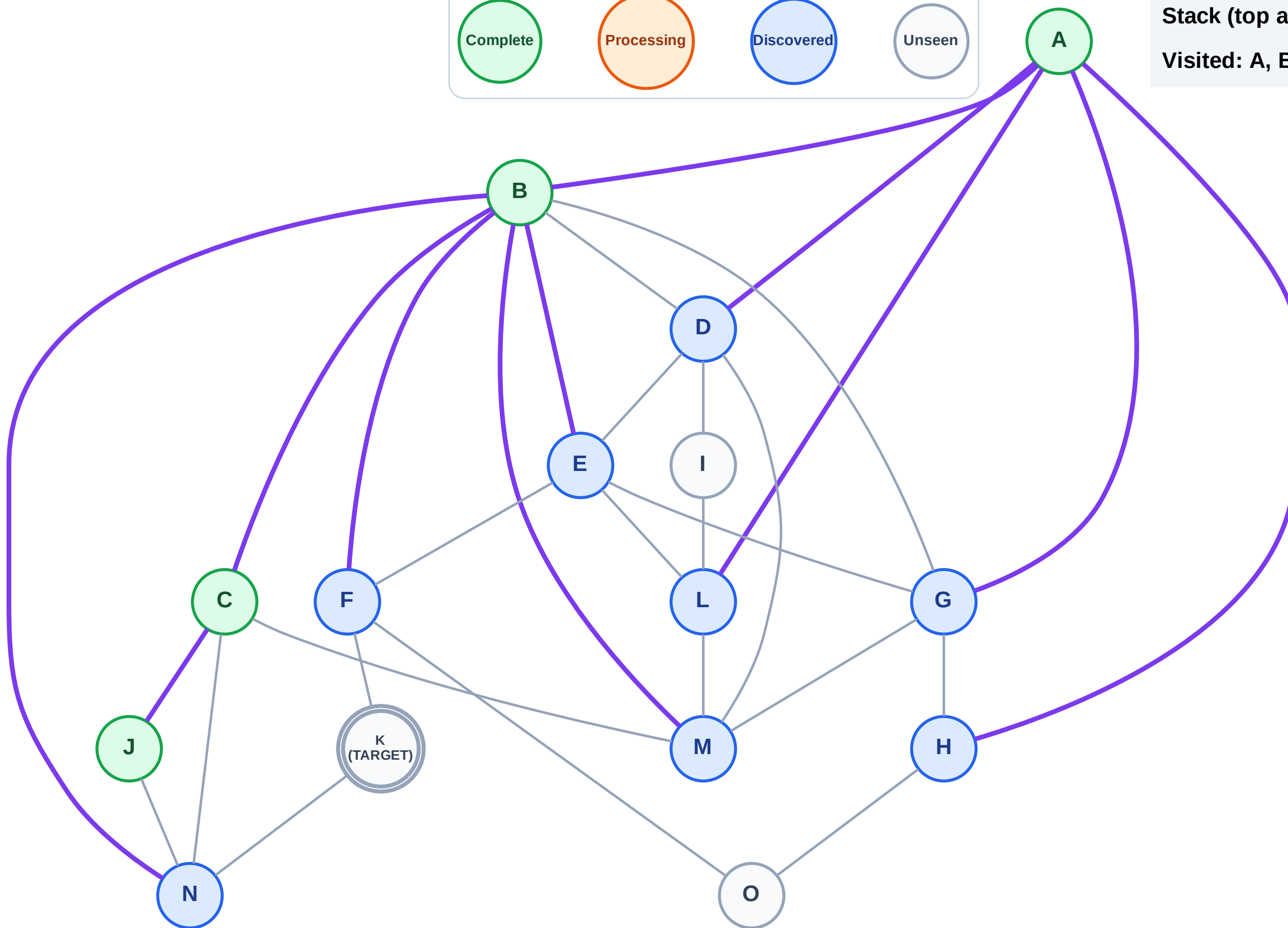
Step 9: No new neighbors from J

## Legend



Stack (top at right): L | H | G | D | N | M | F | E

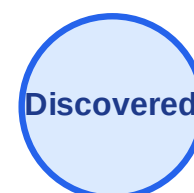
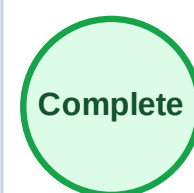
Visited: A, B, C, J



# DFS Traversal

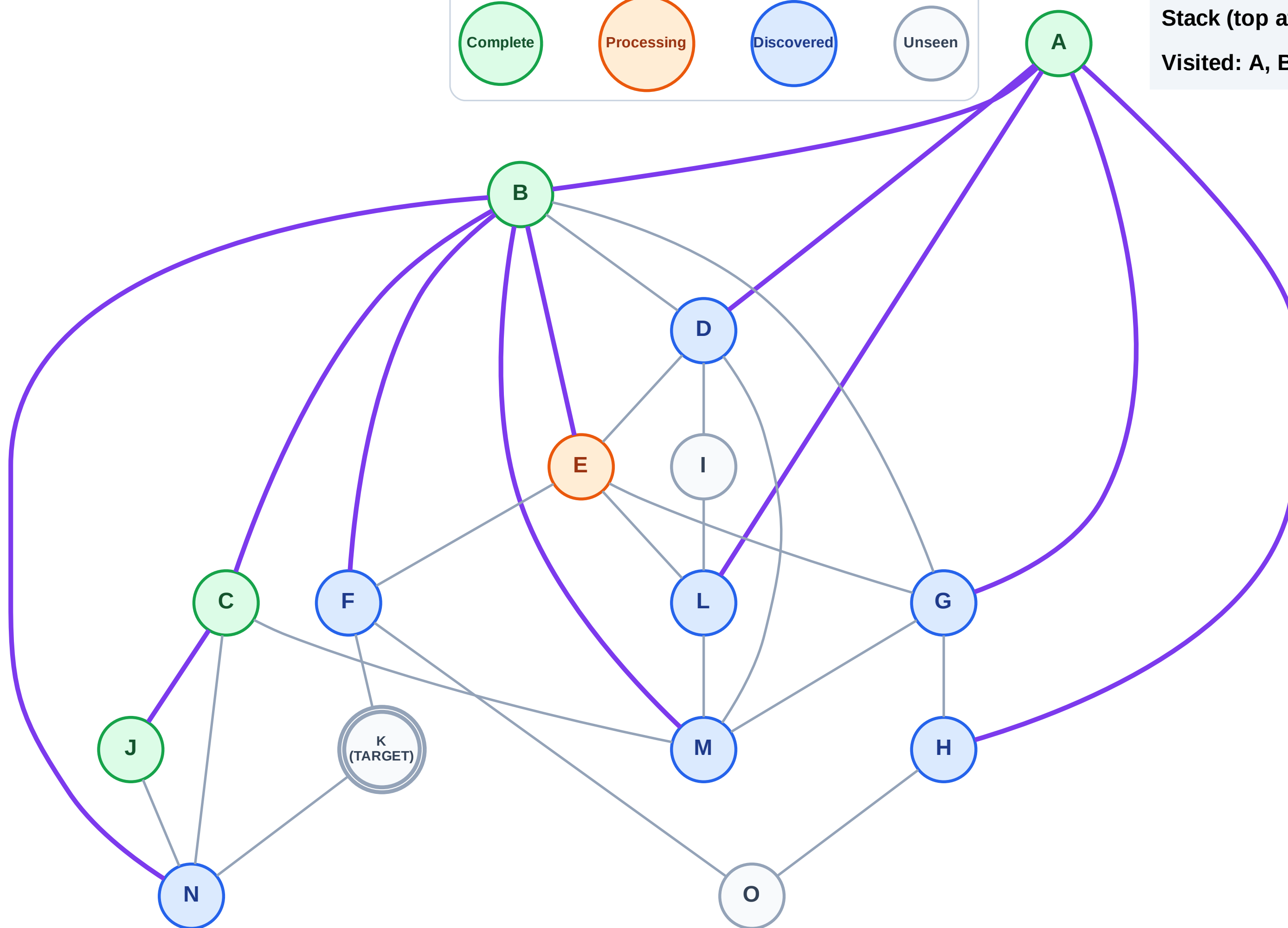
Step 10: Process node E

## Legend



Stack (top at right): L | H | G | D | N | M | F

Visited: A, B, C, J



# DFS Traversal

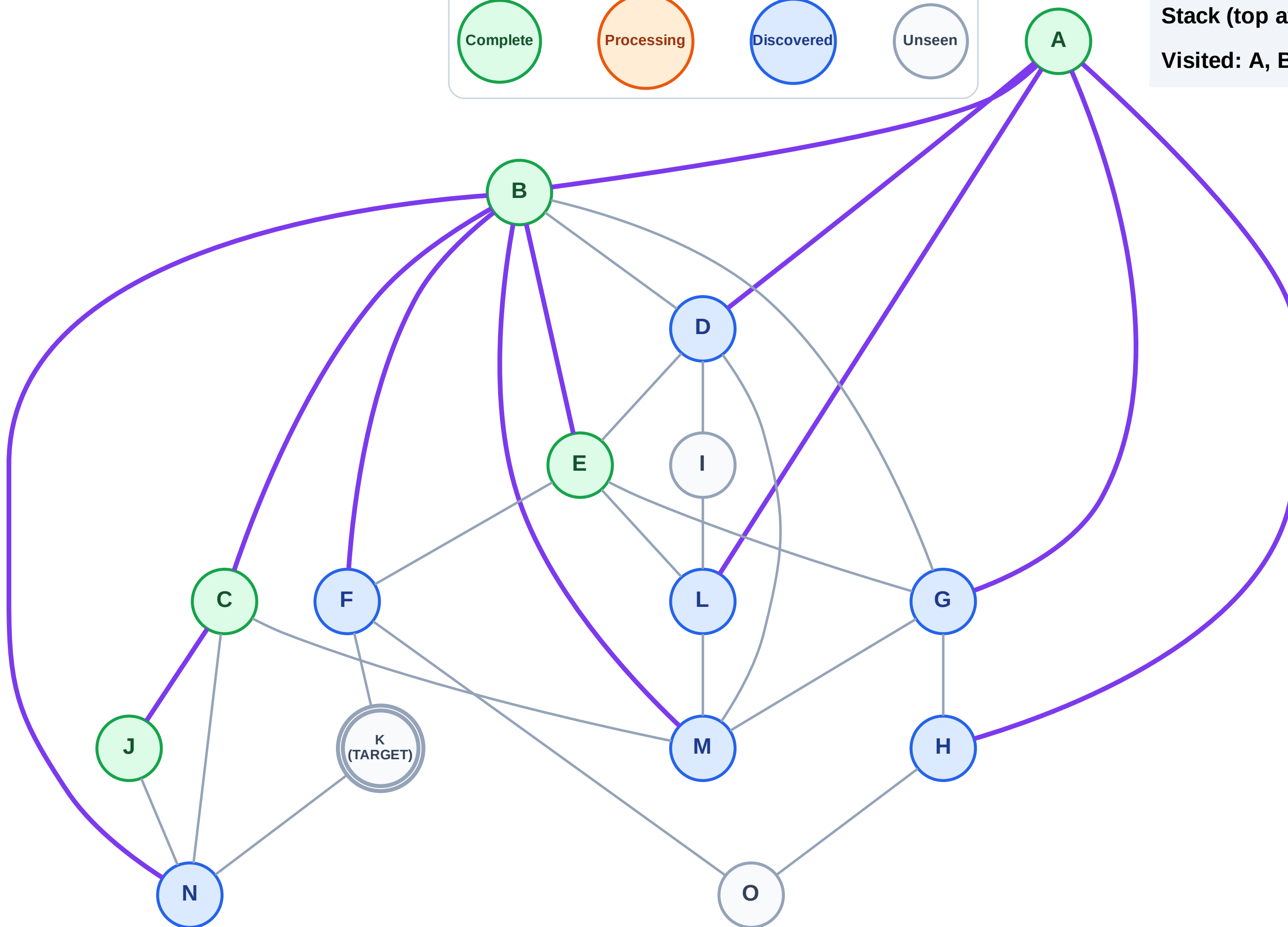
Step 11: No new neighbors from E

Legend



Stack (top at right): L | H | G | D | N | M | F

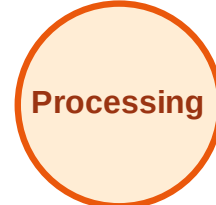
Visited: A, B, C, E, J



# DFS Traversal

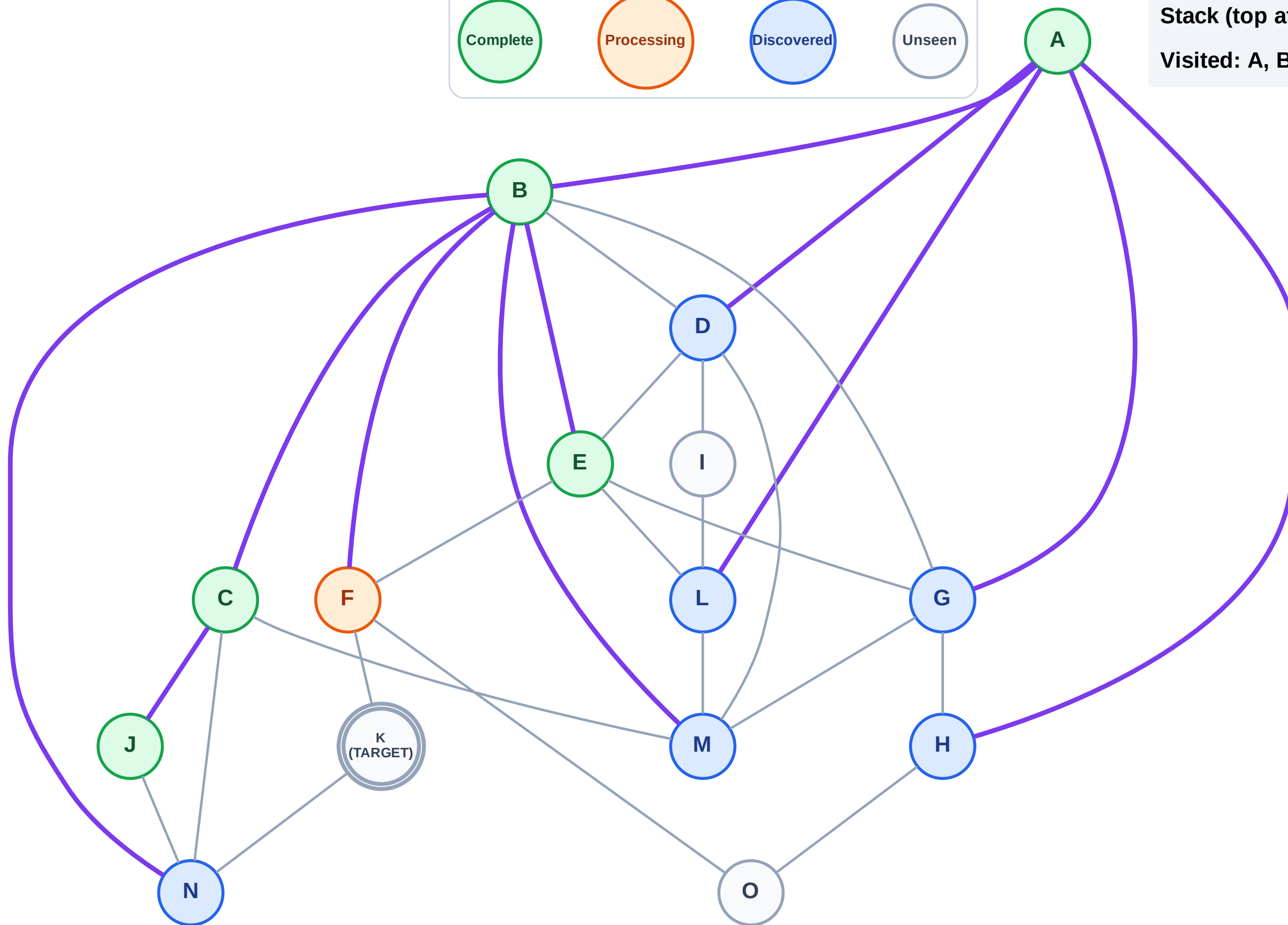
Step 12: Process node F

## Legend



Stack (top at right): L | H | G | D | N | M

Visited: A, B, C, E, J



# DFS Traversal

Step 13: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → F → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): L | H | G | D | N | M | O

Visited: A, B, C, E, F, J, K

